

Chapter 20

Electrochemistry

Electrochemistry

电化学

Electrochemistry is the study of the relationships between **electricity** and **chemical reactions**.

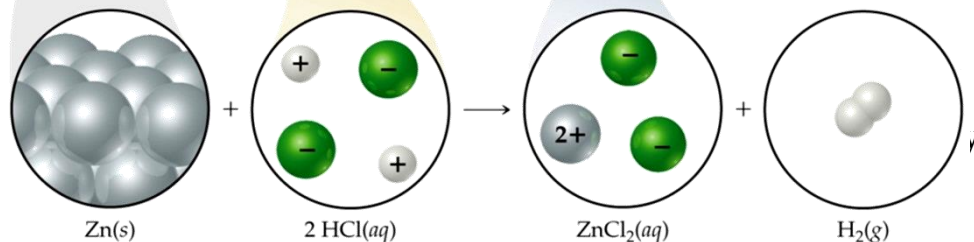
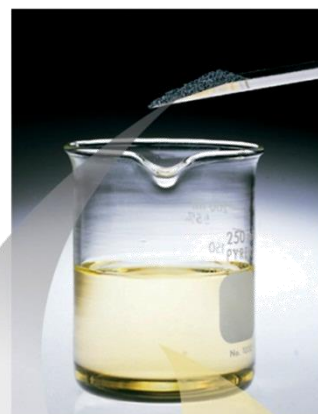
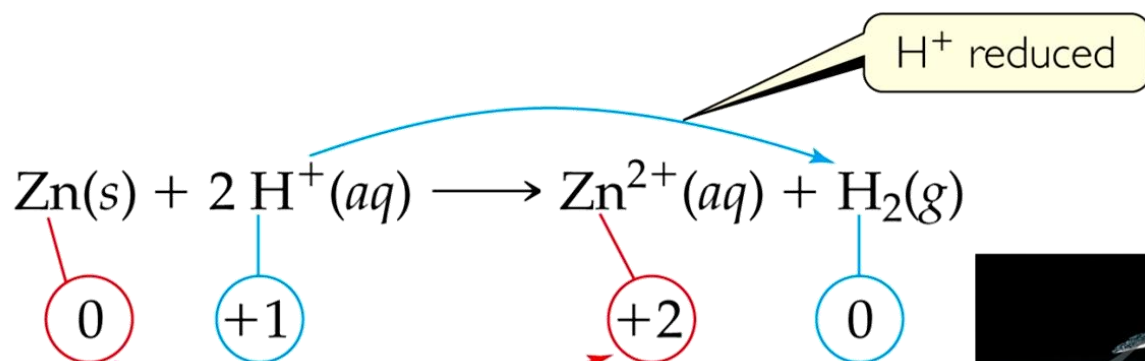
In **electrochemical reactions**, electrons are transferred from one species to another.

电化学主要是研究电能和化学能之间的互相转化以及转化过程中相关规律的科学

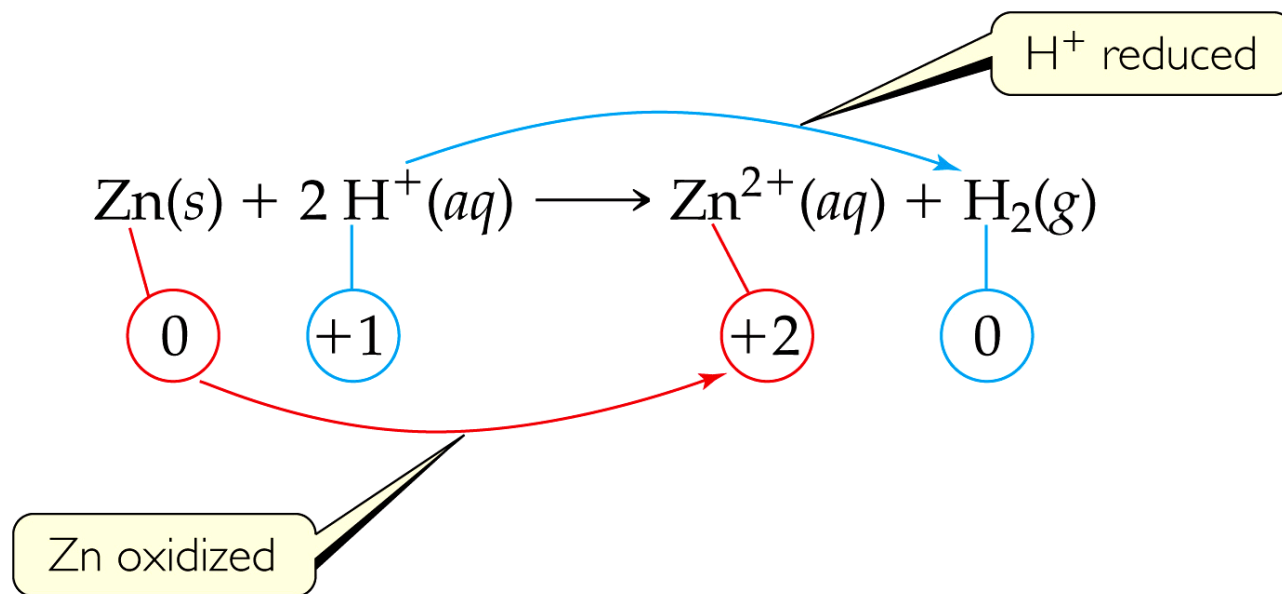


Oxidation Numbers

In order to keep track of what loses electrons and what gains them, we assign **oxidation numbers**.

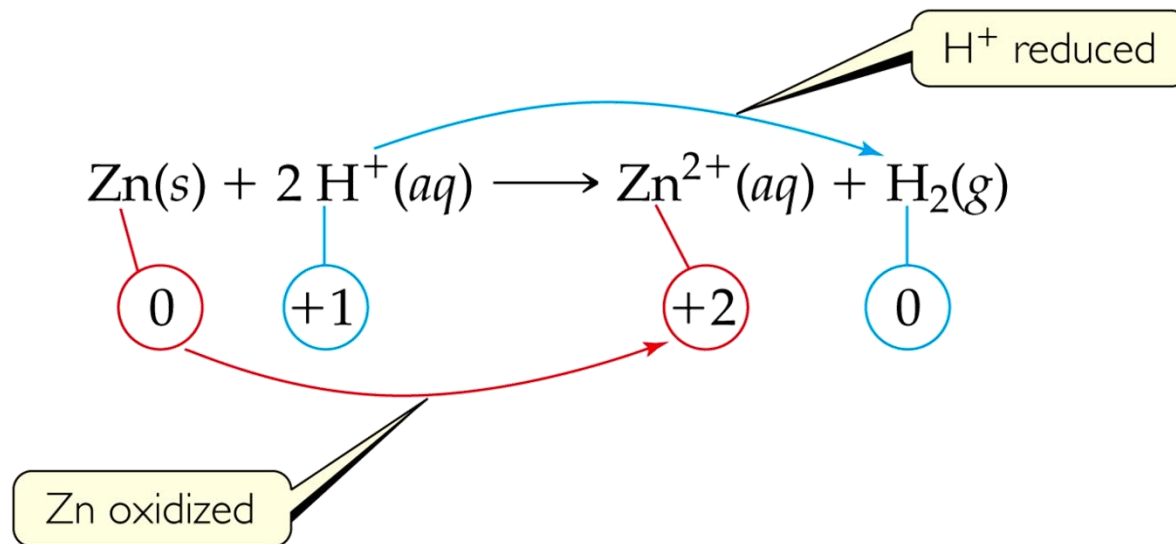


Oxidation and Reduction



- A species is **oxidized** when it loses electrons.
 - Here, zinc loses two electrons to go from neutral zinc metal to the Zn^{2+} ion.
- A species is **reduced** when it gains electrons.
 - Here, each of the H^+ gains an electron, and they combine to form H_2 .

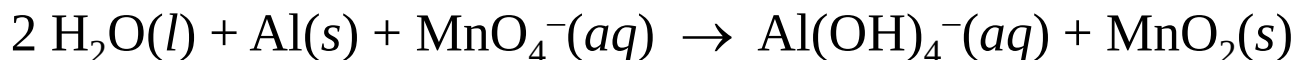
Oxidant and Reductant



- What is reduced is the **oxidizing agent (oxidant)**.
 - H^+ oxidizes Zn by taking electrons from it.
- What is oxidized is the **reducing agent (reductant)**.
 - Zn reduces H^+ by giving it electrons.

Practice Exercise

Identify the oxidizing and reducing agents in the reaction



Electrochemistry



Synopsis of Assigning Oxidation Numbers

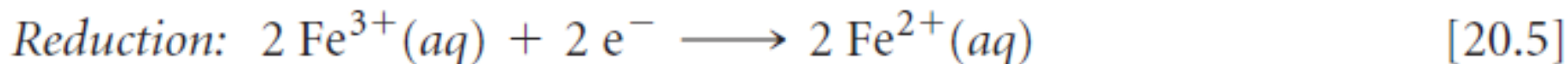
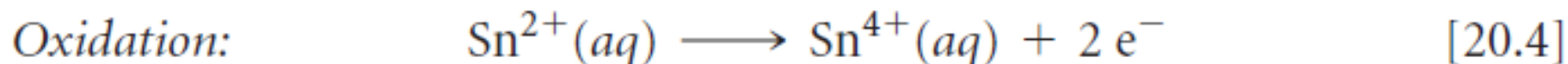
1. Element = 0
2. Monatomic ion = charge
3. F: -1
4. O: -2 (unless peroxide = -1)
5. H: $+1$ (unless a metal hydride = -1)
6. The sum of the oxidation numbers equals the overall charge (0 in a compound).

Balancing Oxidation-Reduction Equations

In electrochemistry, we usually consider the oxidation and reduction process separately since they occur at different electrodes.



Half-reactions:



Whenever we balance a chemical equation, we must obey the law of conservation of mass: The amount of each element must be the same on both sides of the equation. (**Atoms are neither created nor destroyed in any chemical reaction.**)

For oxidation-reduction reactions :

Electrons are neither created nor destroyed in any chemical reaction.

Balancing Redox Equations: The Half-Reactions Method (a Synopsis)

In the overall redox reaction, the number of electrons lost in the oxidation half-reaction must equal the number of electrons gained in the reduction half-reaction.

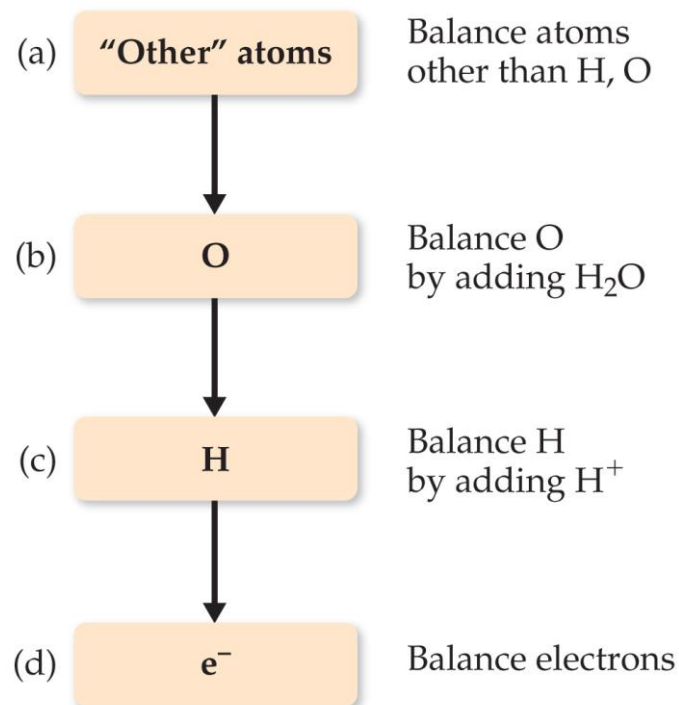
In the half-reaction method, we usually begin with a skeleton (骨架) ionic equation showing only the substances undergoing oxidation and reduction.

We will find that H^+ (for acidic solutions), OH^- (for basic solutions), and H_2O are often involved as reactants or products in redox reactions. Unless H^+ , OH^- , or H_2O is being oxidized or reduced, these species do not appear in the skeleton equation.

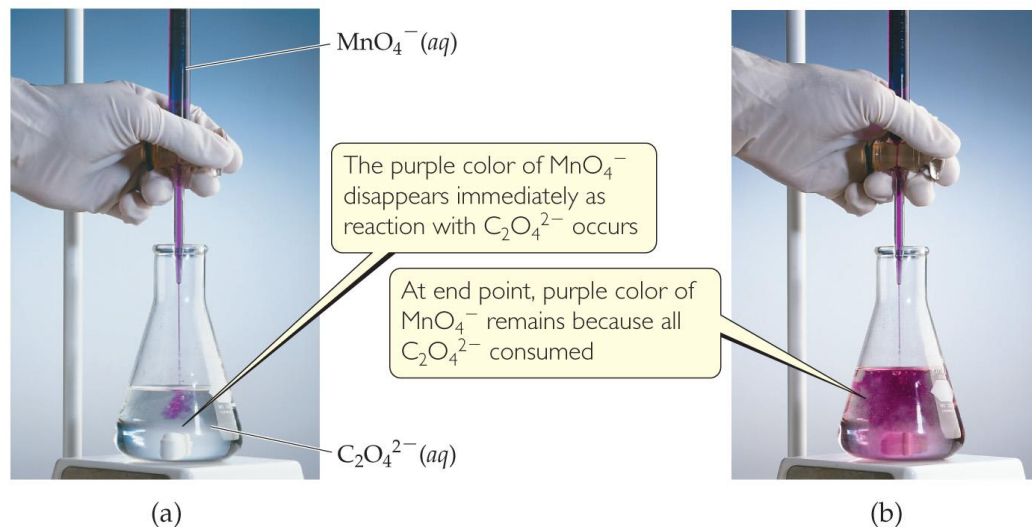


Balancing Redox Equations: The Half-Reactions Method (a Synopsis)

- 1) **Make** two half-reactions (oxidation and reduction).
- 2) **Balance** atoms **other than** O and H. *Then, balance O and H using H_2O/H^+ . (acidic)*
- 3) **Add electrons** to balance charges.
- 4) **Multiply** by common factor to make electrons in half-reactions equal.
- 5) **Add** the half-reactions, subtracting things that appear on both sides.
- 6) **Converting** H^+ to OH^- if it is under basic condition.
- 7) **Double-check** atoms and charges balance!



The Half-Reaction Method 半反应法



Consider the reaction between MnO_4^- and $\text{C}_2\text{O}_4^{2-}$:

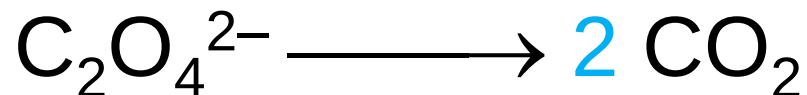


- Assigning oxidation numbers shows that **Mn** is reduced (+7 $\rightarrow$ +2) and **C** is oxidized (+3 $\rightarrow$ +4).

Oxidation Half-Reaction



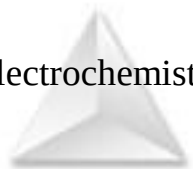
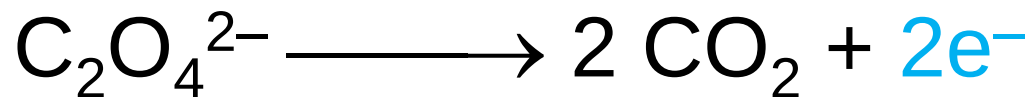
To balance the carbon, we add a coefficient of 2:



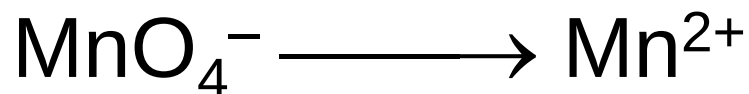
Oxidation Half-Reaction



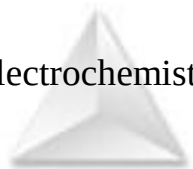
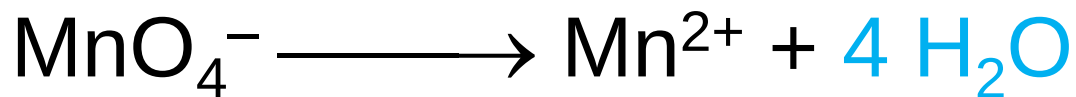
The oxygen is now balanced as well. To balance the charge, we must add two electrons to the right side:



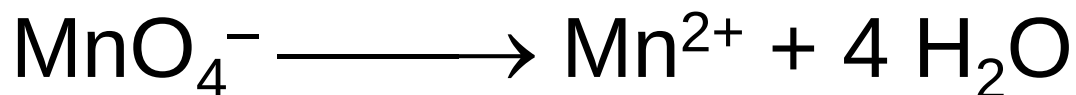
Reduction Half-Reaction



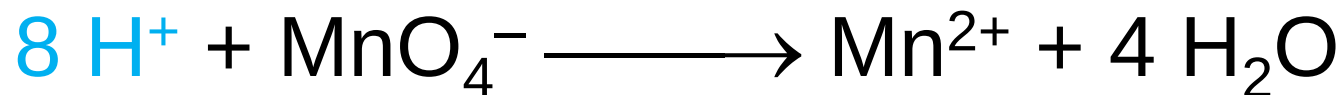
The manganese is balanced; to balance the oxygen, we must add four waters to the right side:



Reduction Half-Reaction



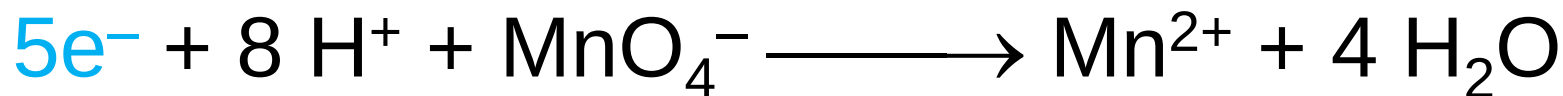
To balance the **hydrogen**, we add 8H^+ to the left side:



Reduction Half-Reaction

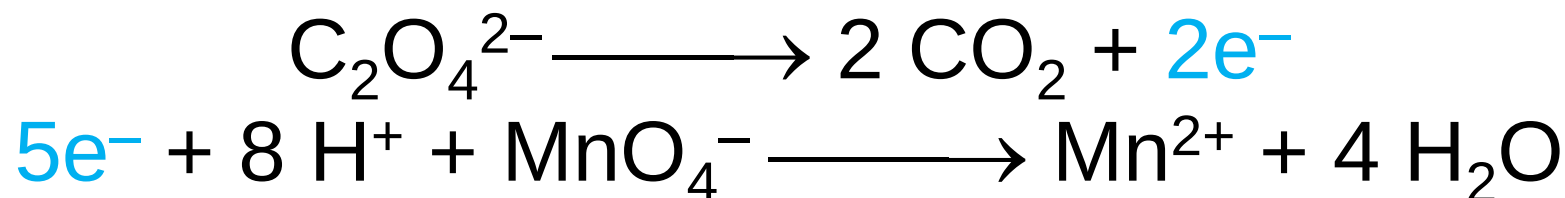


To balance the charge, we add 5e^- to the left side:



Combining the Half-Reactions

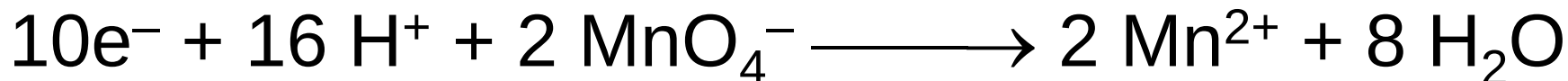
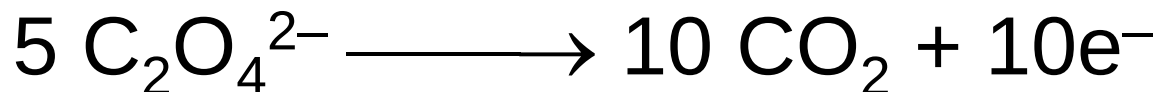
Now we combine the two half-reactions together:



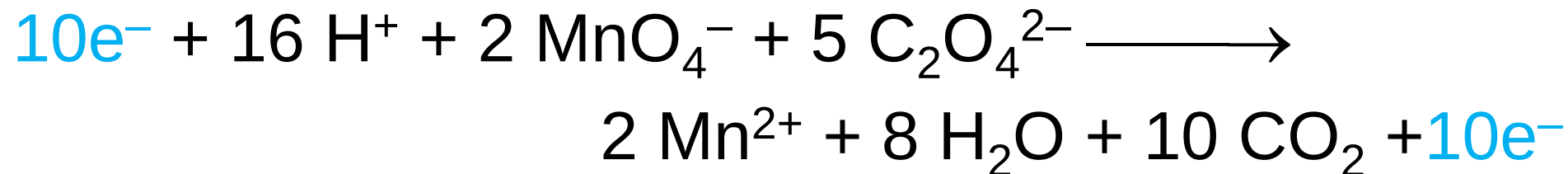
To make the number of electrons equal on each side, we will multiply the first reaction by 5 and the second by 2:



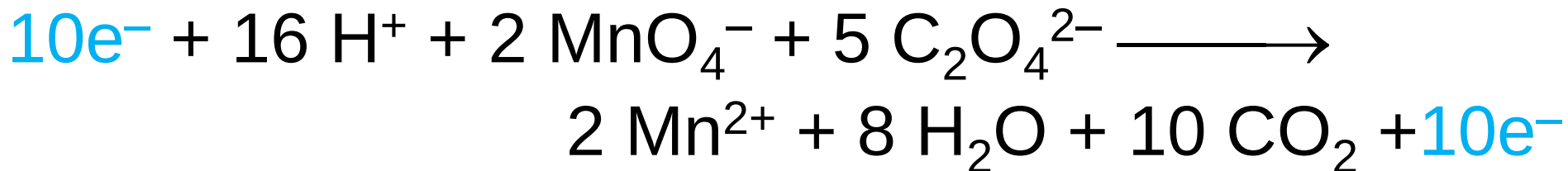
Combining the Half-Reactions



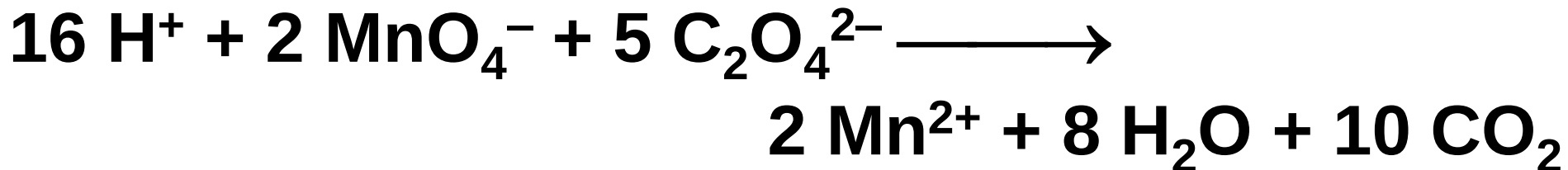
When we **add** these together, we get



Combining the Half-Reactions



The only thing that appears on both sides is the electrons. Subtracting them, we are left with



(**Verify** that the equation is balanced by counting atoms and charges on each side of the equation.)



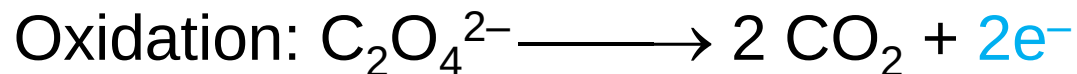
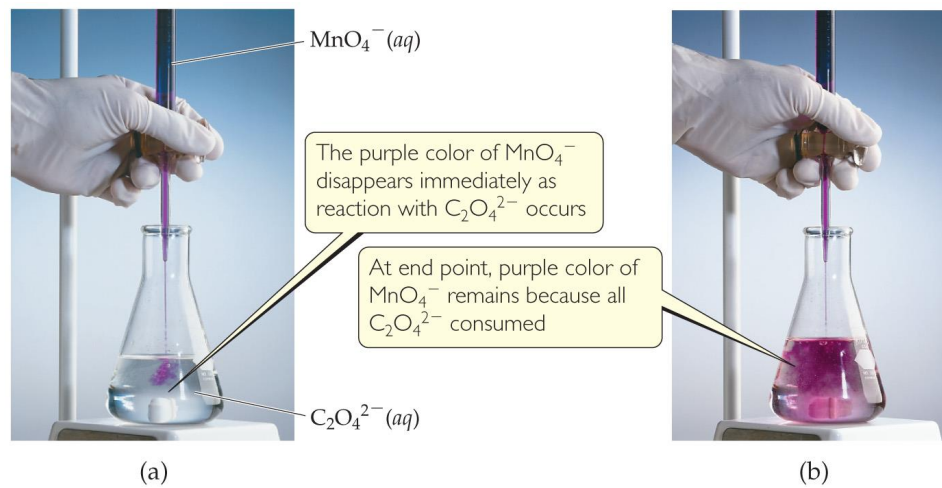
Complete and balance the following equation:



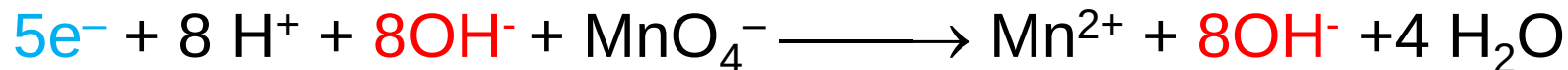
Balancing in Basic Solution

- A reaction that occurs in basic solution can be balanced as if it occurred in acid.
- Once the equation is balanced, add OH^- to each side to “neutralize” the H^+ in the equation and create water in its place.
- If this produces water on both sides, subtract water from each side so it appears on only one side of the equation.

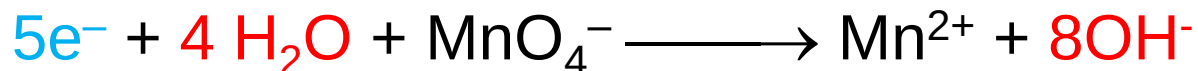




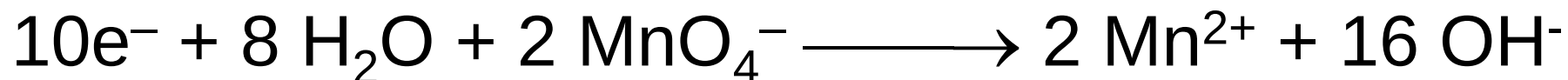
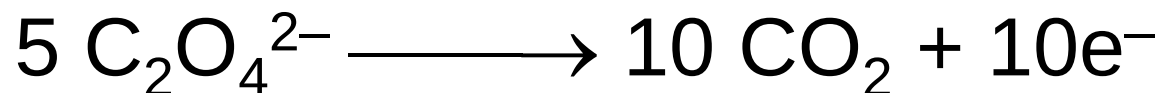
Under basic condition, H^+ is neutralized by OH^-



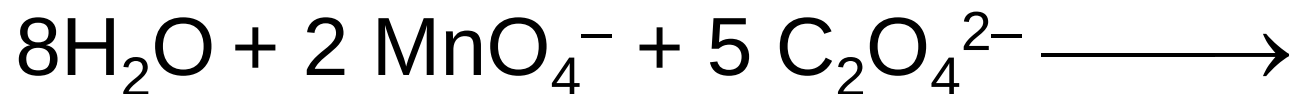
Simplify:



Combining the Half-Reactions



When we **add** these together, we get



SAMPLE

EXERCISE 20.3

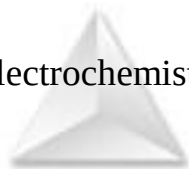
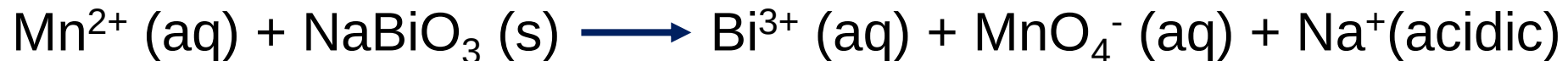
Balancing Redox Equations in Basic Solution

Complete and balance this equation for a redox reaction that takes place in basic solution:



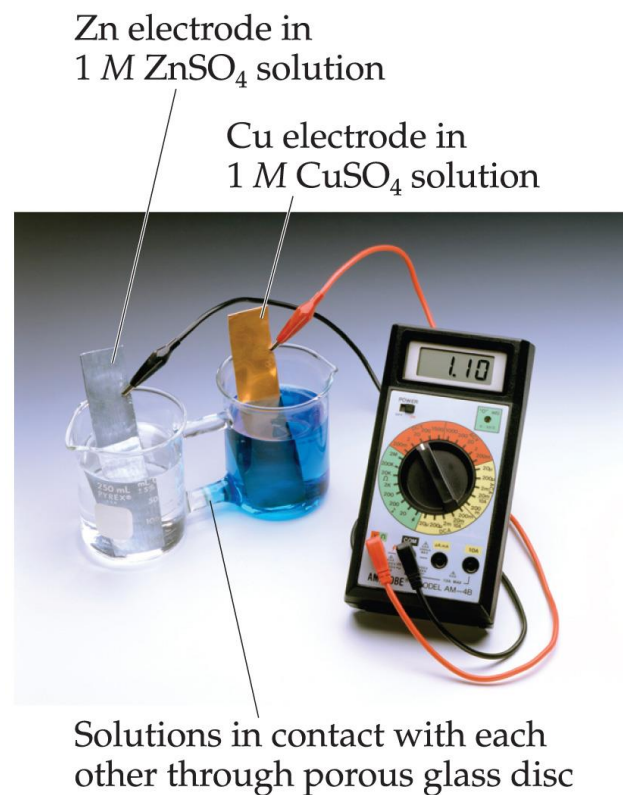
Practice Exercise

Complete and balance the following equations:



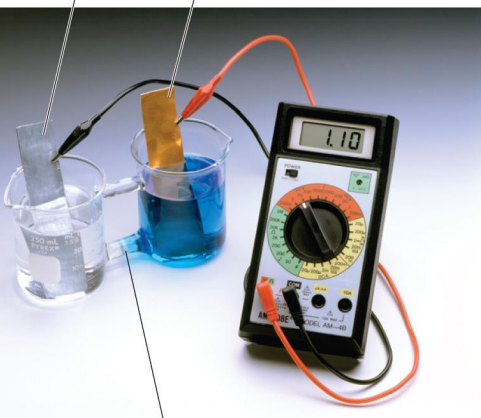
Voltaic Cells 原电池/伏打电池

- In spontaneous redox reactions, **electrons** are transferred and **energy** is released.
- That energy can do work if the electrons flow through an **external** device.
- This is a **voltaic cell**.

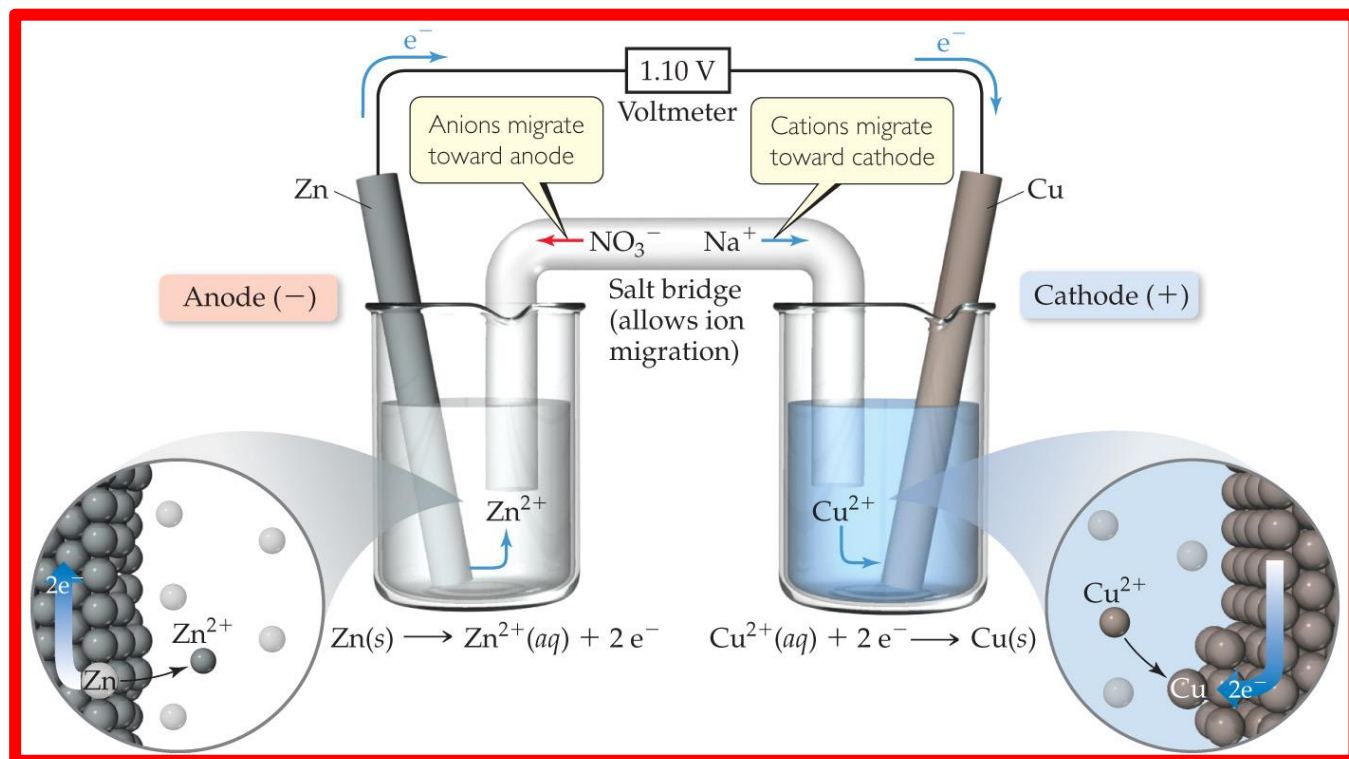


Zn electrode in
1 M ZnSO_4 solution

Cu electrode in
1 M CuSO_4 solution



Solutions in contact with each
other through porous glass disc

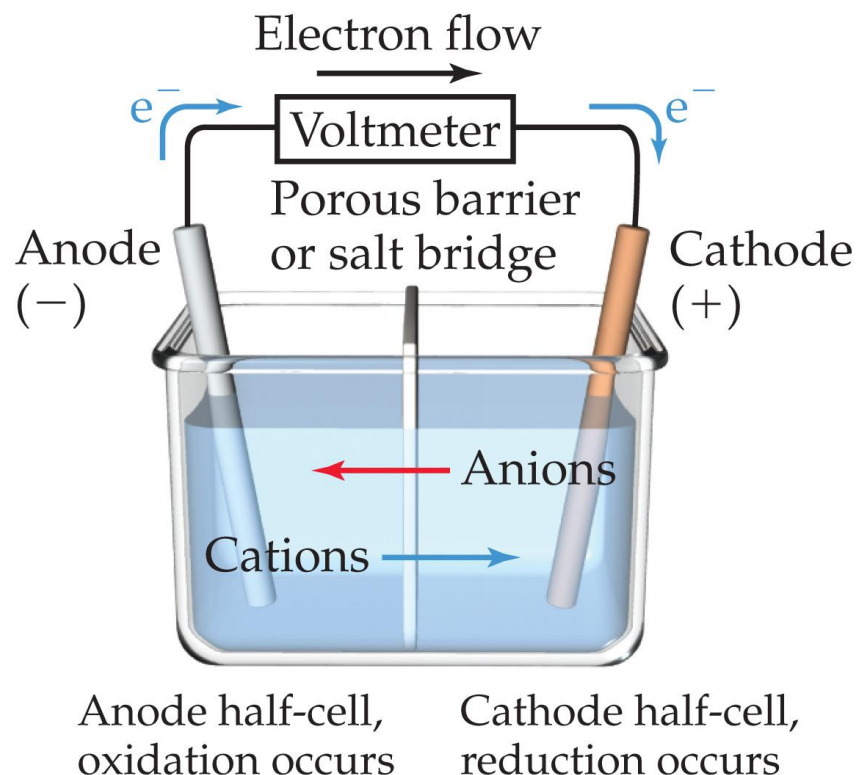


- The **oxidation** occurs at the **anode**. 阳极
- The **reduction** occurs at the **cathode**. 阴极
- When electrons flow, **charges** aren't balanced. So, a salt bridge 盐桥, usually a U-shaped tube that contains a salt/agar solution, **is used to keep the charges balanced**.

Electrochemistry

Voltaic Cells 原电池

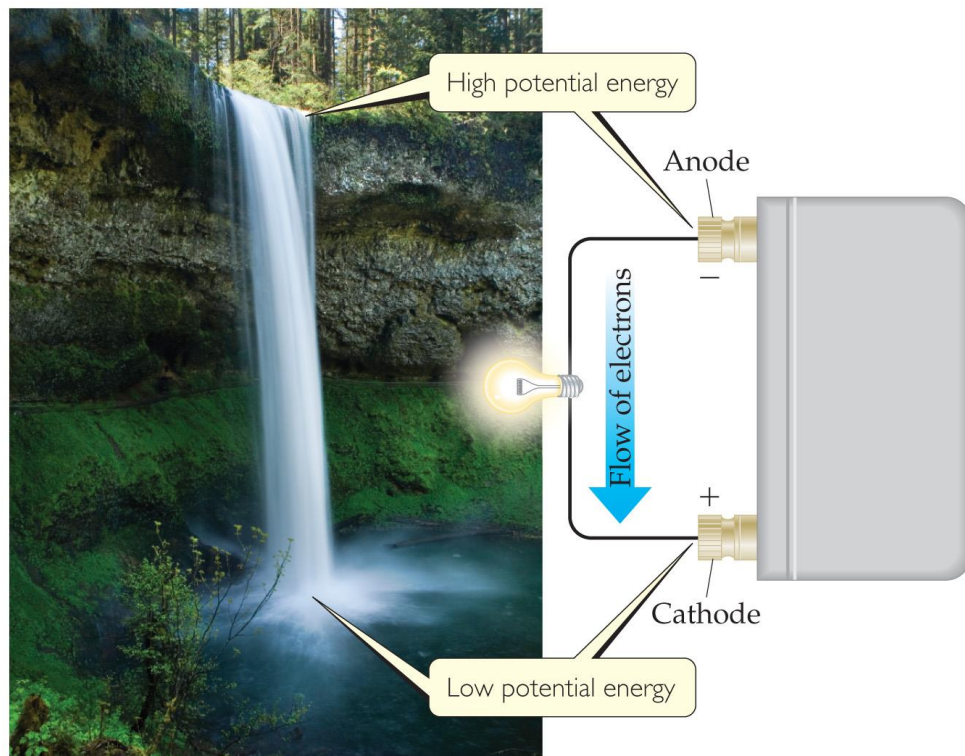
- In the cell, **electrons** leave the anode and flow through the wire to the cathode.
- Cations are formed in the anode compartment.
- As the electrons reach the cathode, cations in solution are attracted to the now negative cathode.
- The cations gain electrons and are deposited as metal on the cathode.



Electrochemistry

Electromotive Force (emf) 电动势

- Water flows spontaneously one way in a waterfall.
- Comparably, electrons flow spontaneously one way in a redox reaction, from high to low potential energy.



Electromotive Force (emf)

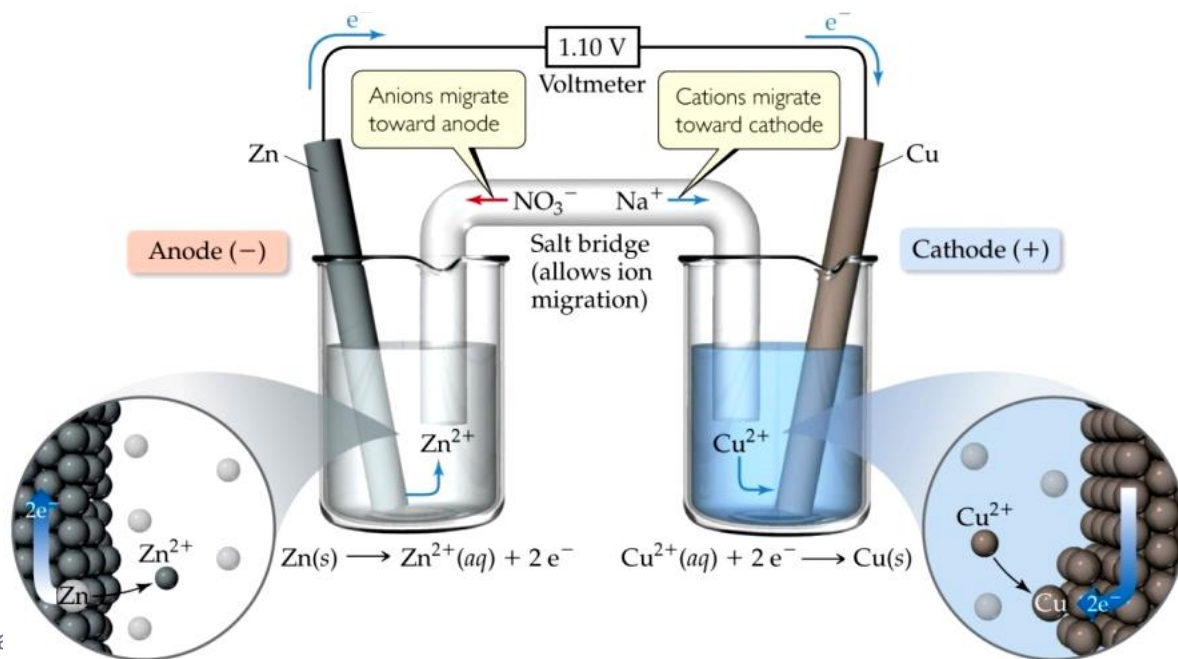
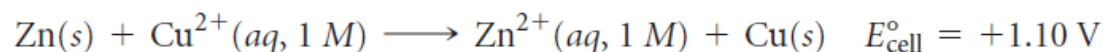
- The potential difference between the anode and cathode in a cell is called the **electromotive force (emf)**.
- It is also called the **cell potential (电池电位)** and is designated E_{cell} .
- It is measured in **volts (V)**. One volt is one joule per coulomb ($1 \text{ V} = 1 \text{ J/C}$).



Electromotive Force (emf)

standard conditions include **1 M concentrations** for reactants and products in solution and **1 atm pressure** for gaseous reactants and products. The cell potential under standard conditions is called either the **standard cell potential** or **standard emf** and is denoted

$$E_{\text{cell}}^{\circ}$$

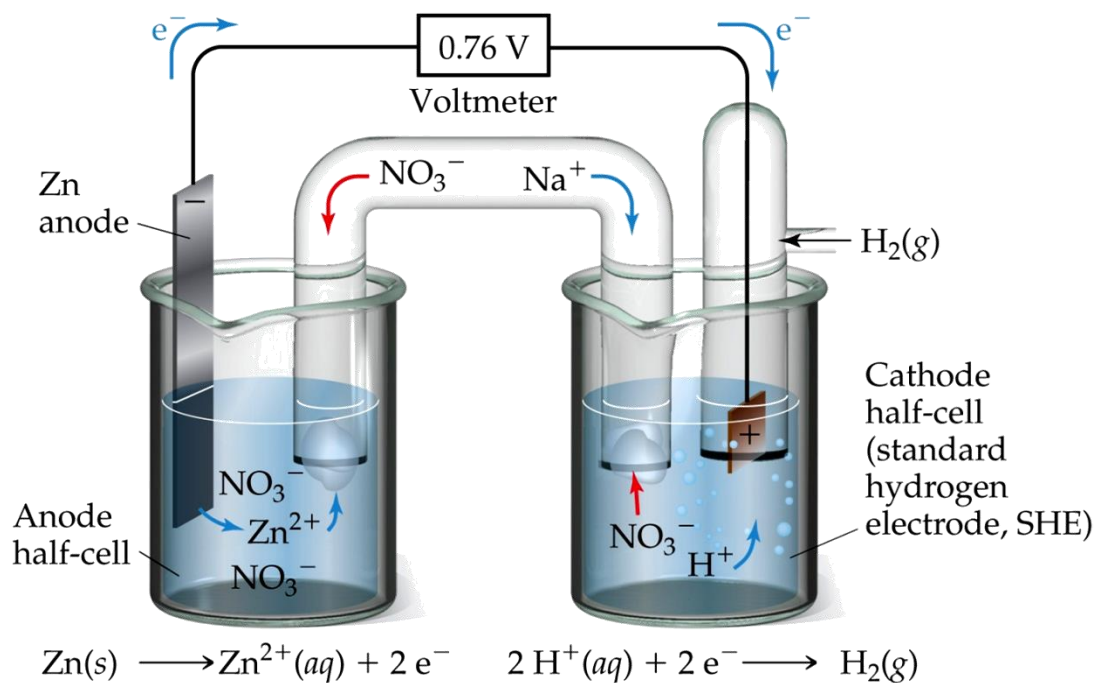


Electrochemistry

Standard Potentials

The cell potential at standard conditions can be found through this equation:

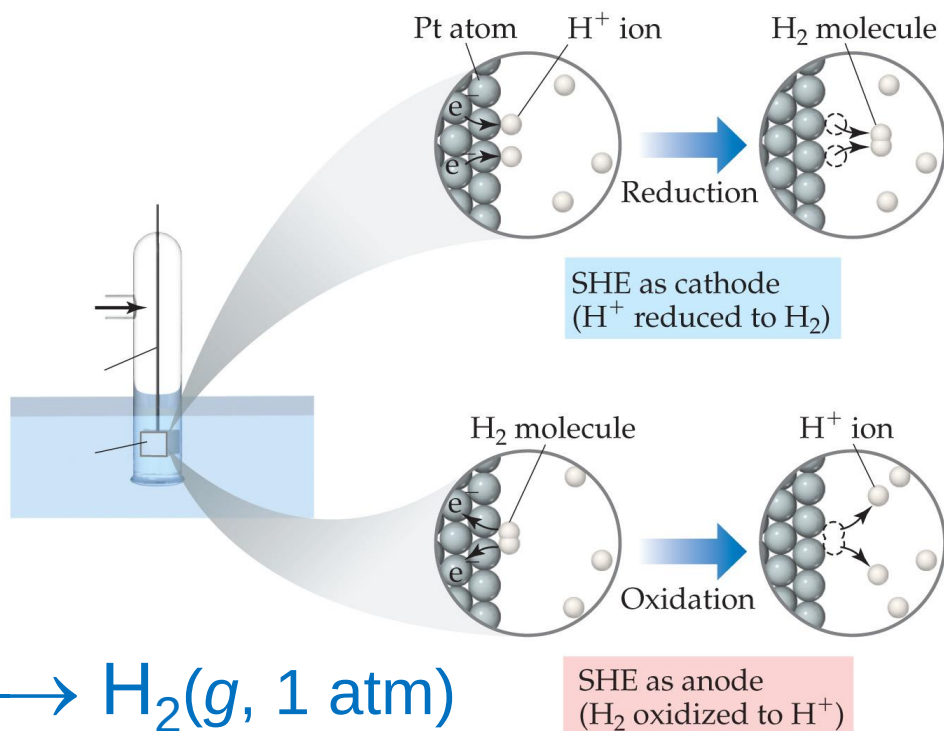
$$E_{\text{cell}}^{\circ} = E^{\circ}(\text{cathode}) - E^{\circ}(\text{anode})$$



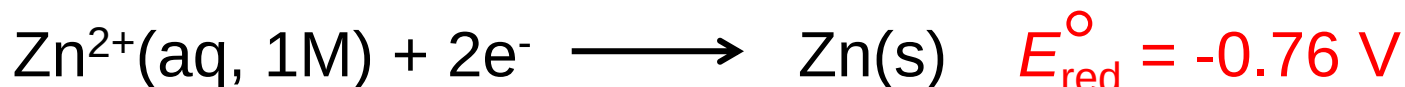
Whenever we assign an electrical potential to a half-reaction, we write the reaction as a reduction.

Standard Hydrogen Electrode 标准氢电极

- Their reference is called the **standard hydrogen electrode (SHE)**.
- By definition as the standard, the reduction potential for **hydrogen** is **0 V**:



Therefore,



Electrochemistry

Standard Reduction Potentials

标准还原电位

Table 20.1 Standard Reduction Potentials in Water at 25 °C

$E^\circ_{\text{red}}(\text{V})$	Reduction Half-Reaction
+2.87	$\text{F}_2(\text{g}) + 2 \text{e}^- \longrightarrow 2 \text{F}^-(\text{aq})$
+1.51	$\text{MnO}_4^-(\text{aq}) + 8 \text{H}^+(\text{aq}) + 5 \text{e}^- \longrightarrow \text{Mn}^{2+}(\text{aq}) + 4 \text{H}_2\text{O}(\text{l})$
+1.36	$\text{Cl}_2(\text{g}) + 2 \text{e}^- \longrightarrow 2 \text{Cl}^-(\text{aq})$
+1.33	$\text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 14 \text{H}^+(\text{aq}) + 6 \text{e}^- \longrightarrow 2 \text{Cr}^{3+}(\text{aq}) + 7 \text{H}_2\text{O}(\text{l})$
+1.23	$\text{O}_2(\text{g}) + 4 \text{H}^+(\text{aq}) + 4 \text{e}^- \longrightarrow 2 \text{H}_2\text{O}(\text{l})$
+1.06	$\text{Br}_2(\text{l}) + 2 \text{e}^- \longrightarrow 2 \text{Br}^-(\text{aq})$
+0.96	$\text{NO}_3^-(\text{aq}) + 4 \text{H}^+(\text{aq}) + 3 \text{e}^- \longrightarrow \text{NO}(\text{g}) + 2 \text{H}_2\text{O}(\text{l})$
+0.80	$\text{Ag}^+(\text{aq}) + \text{e}^- \longrightarrow \text{Ag}(\text{s})$
+0.77	$\text{Fe}^{3+}(\text{aq}) + \text{e}^- \longrightarrow \text{Fe}^{2+}(\text{aq})$
+0.68	$\text{O}_2(\text{g}) + 2 \text{H}^+(\text{aq}) + 2 \text{e}^- \longrightarrow \text{H}_2\text{O}_2(\text{aq})$
+0.59	$\text{MnO}_4^-(\text{aq}) + 2 \text{H}_2\text{O}(\text{l}) + 3 \text{e}^- \longrightarrow \text{MnO}_2(\text{s}) + 4 \text{OH}^-(\text{aq})$
+0.54	$\text{I}_2(\text{s}) + 2 \text{e}^- \longrightarrow 2 \text{I}^-(\text{aq})$
+0.40	$\text{O}_2(\text{g}) + 2 \text{H}_2\text{O}(\text{l}) + 4 \text{e}^- \longrightarrow 4 \text{OH}^-(\text{aq})$
+0.34	$\text{Cu}^{2+}(\text{aq}) + 2 \text{e}^- \longrightarrow \text{Cu}(\text{s})$
0 [defined]	$2 \text{H}^+(\text{aq}) + 2 \text{e}^- \longrightarrow \text{H}_2(\text{g})$
-0.28	$\text{Ni}^{2+}(\text{aq}) + 2 \text{e}^- \longrightarrow \text{Ni}(\text{s})$
-0.44	$\text{Fe}^{2+}(\text{aq}) + 2 \text{e}^- \longrightarrow \text{Fe}(\text{s})$
-0.76	$\text{Zn}^{2+}(\text{aq}) + 2 \text{e}^- \longrightarrow \text{Zn}(\text{s})$
-0.83	$2 \text{H}_2\text{O}(\text{l}) + 2 \text{e}^- \longrightarrow \text{H}_2(\text{g}) + 2 \text{OH}^-(\text{aq})$
-1.66	$\text{Al}^{3+}(\text{aq}) + 3 \text{e}^- \longrightarrow \text{Al}(\text{s})$
-2.71	$\text{Na}^+(\text{aq}) + \text{e}^- \longrightarrow \text{Na}(\text{s})$
-3.05	$\text{Li}^+(\text{aq}) + \text{e}^- \longrightarrow \text{Li}(\text{s})$

- Reduction potentials for many electrodes have been measured and tabulated.
- The values are compared to the reduction of hydrogen as a standard.

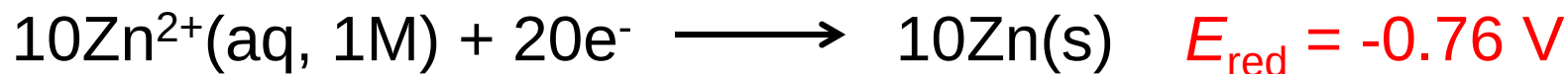
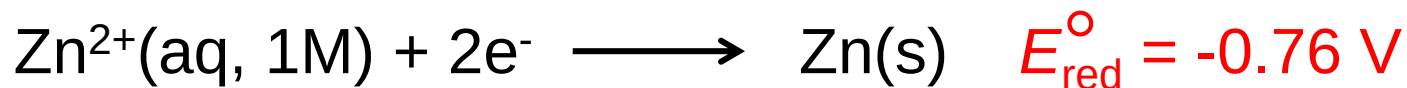


Standard Cell Potentials

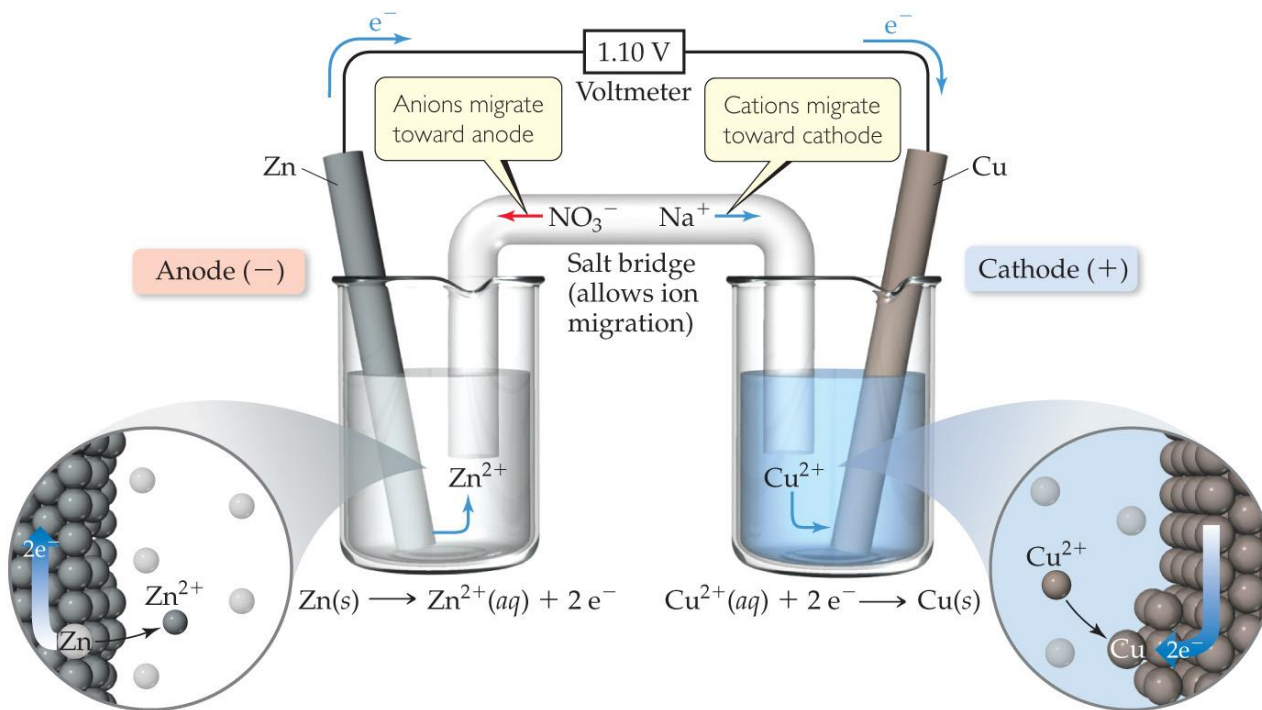
标准电池电位

$$E_{\text{cell}}^{\circ} = E_{\text{red}}^{\circ} (\text{cathode}) - E_{\text{red}}^{\circ} (\text{anode})$$

Because **cell potential** is based on the potential energy per unit of charge, it is an **intensive** property. (1 V = 1 J/C)



Cell Potentials

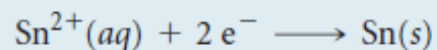
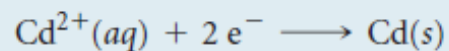


- For the anode in this cell, $E^\circ_{\text{red}} = -0.76 \text{ V}$
- For the cathode, $E^\circ_{\text{red}} = +0.34 \text{ V}$
- So, for the cell, $E^\circ_{\text{cell}} = E^\circ_{\text{red}} (\text{cathode}) - E^\circ_{\text{red}} (\text{anode})$

$$= +0.34 \text{ V} - (-0.76 \text{ V}) = \textcolor{red}{+1.10 \text{ V}} \text{ (always positive)}$$

SAMPLE EXERCISE 20.7 Determining Half-Reactions at Electrodes and Calculating Cell Potentials

A voltaic cell is based on the two standard half-reactions



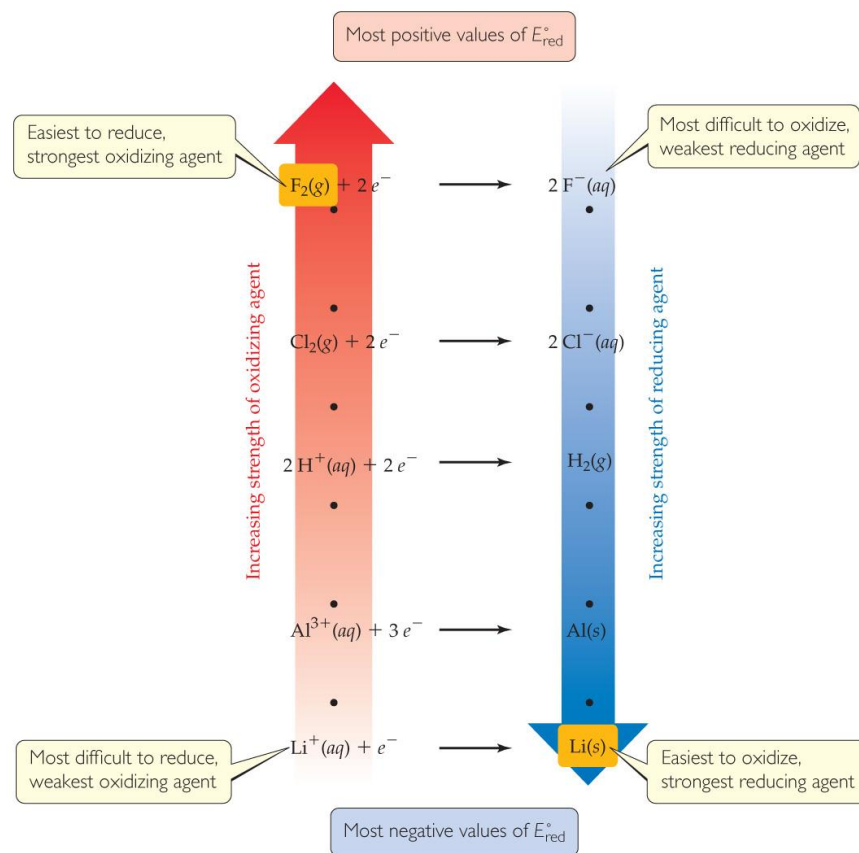
Use data in Appendix E to determine (a) which half-reaction occurs at the cathode and which occurs at the anode and (b) the standard cell potential.



Oxidizing and Reducing Agents

氧化剂/还原剂

- The more positive the value of E°_{red} , the greater the tendency for reduction under standard conditions.
- The strongest oxidizers have the most positive reduction potentials.
- The strongest reducers have the most negative reduction potentials.



Electrochemistry

SAMPLE EXERCISE 20.8

Determining Relative Strengths of Oxidizing Agents

Using Table 20.1, rank the following ions in order of increasing strength as oxidizing agents: $\text{NO}_3^-(aq)$, $\text{Ag}^+(aq)$, $\text{Cr}_2\text{O}_7^{2-}(aq)$.

Table 20.1 Standard Reduction Potentials in Water at 25 °C

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+1.23	$\text{O}_2(\text{g}) + 4 \text{H}^+(\text{aq}) + 4 \text{e}^- \longrightarrow 2 \text{H}_2\text{O}(\text{l})$
+1.06	$\text{Br}_2(\text{l}) + 2 \text{e}^- \longrightarrow 2 \text{Br}^-(\text{aq})$
+0.96	$\text{NO}_3^-(\text{aq}) + 4 \text{H}^+(\text{aq}) + 3 \text{e}^- \longrightarrow \text{NO}(\text{g}) + 2 \text{H}_2\text{O}(\text{l})$
+0.80	$\text{Ag}^+(\text{aq}) + \text{e}^- \longrightarrow \text{Ag}(\text{s})$
+0.77	$\text{Fe}^{3+}(\text{aq}) + \text{e}^- \longrightarrow \text{Fe}^{2+}(\text{aq})$
+0.68	$\text{O}_2(\text{g}) + 2 \text{H}^+(\text{aq}) + 2 \text{e}^- \longrightarrow \text{H}_2\text{O}_2(\text{aq})$
+0.59	$\text{MnO}_4^-(\text{aq}) + 2 \text{H}_2\text{O}(\text{l}) + 3 \text{e}^- \longrightarrow \text{MnO}_2(\text{s}) + 4 \text{OH}^-(\text{aq})$
+0.54	$\text{I}_2(\text{s}) + 2 \text{e}^- \longrightarrow 2 \text{I}^-(\text{aq})$
+0.40	$\text{O}_2(\text{g}) + 2 \text{H}_2\text{O}(\text{l}) + 4 \text{e}^- \longrightarrow 4 \text{OH}^-(\text{aq})$
+0.34	$\text{Cu}^{2+}(\text{aq}) + 2 \text{e}^- \longrightarrow \text{Cu}(\text{s})$
0 [defined]	$2 \text{H}^+(\text{aq}) + 2 \text{e}^- \longrightarrow \text{H}_2(\text{g})$
-0.28	$\text{Ni}^{2+}(\text{aq}) + 2 \text{e}^- \longrightarrow \text{Ni}(\text{s})$
-0.44	$\text{Fe}^{2+}(\text{aq}) + 2 \text{e}^- \longrightarrow \text{Fe}(\text{s})$
-0.76	$\text{Zn}^{2+}(\text{aq}) + 2 \text{e}^- \longrightarrow \text{Zn}(\text{s})$
-0.83	$2 \text{H}_2\text{O}(\text{l}) + 2 \text{e}^- \longrightarrow \text{H}_2(\text{g}) + 2 \text{OH}^-(\text{aq})$
-1.66	$\text{Al}^{3+}(\text{aq}) + 3 \text{e}^- \longrightarrow \text{Al}(\text{s})$
-2.71	$\text{Na}^+(\text{aq}) + \text{e}^- \longrightarrow \text{Na}(\text{s})$
-3.05	$\text{Li}^+(\text{aq}) + \text{e}^- \longrightarrow \text{Li}(\text{s})$

Spontaneity of Redox Reactions

$$E_{\text{cell}}^{\circ} = E_{\text{red}}^{\circ} (\text{cathode}) - E_{\text{red}}^{\circ} (\text{anode}) > 0 \text{ (voltaic)}$$



$$E^{\circ} = E_{\text{red}}^{\circ} (\text{reduction}) - E_{\text{red}}^{\circ} (\text{oxidation})$$

- Spontaneous redox reactions produce a **positive cell potential**, or emf.

We can now make a general statement about the spontaneity of a reaction and its associated emf, E :

***A positive value of E indicates a spontaneous process;
a negative value of E indicates a nonspontaneous process.***





$$E^\circ = E^\circ_{\text{red}}(\text{Ag}^+/\text{Ag}) - E^\circ_{\text{red}}(\text{Ni}^{2+}/\text{Ni})$$

$$= (+0.80 \text{ V}) - (-0.28 \text{ V}) = +1.08 \text{ V}$$

We can use standard reduction potentials to understand the activity series of metals.

Table 20.1 Standard Reduction Potentials in Water at 25 °C

$E^\circ_{\text{red}}(\text{V})$	Reduction Half-Reaction
+2.87	$\text{F}_2(g) + 2 e^- \longrightarrow 2 \text{F}^-(aq)$
+1.51	$\text{MnO}_4^-(aq) + 8 \text{H}^+(aq) + 5 e^- \longrightarrow \text{Mn}^{2+}(aq) + 4 \text{H}_2\text{O}(l)$
+1.36	$\text{Cl}_2(g) + 2 e^- \longrightarrow 2 \text{Cl}^-(aq)$
+1.33	$\text{Cr}_2\text{O}_7^{2-}(aq) + 14 \text{H}^+(aq) + 6 e^- \longrightarrow 2 \text{Cr}^{3+}(aq) + 7 \text{H}_2\text{O}(l)$
+1.23	$\text{O}_2(g) + 4 \text{H}^+(aq) + 4 e^- \longrightarrow 2 \text{H}_2\text{O}(l)$
+1.06	$\text{Br}_2(l) + 2 e^- \longrightarrow 2 \text{Br}^-(aq)$
+0.96	$\text{NO}_3^-(aq) + 4 \text{H}^+(aq) + 3 e^- \longrightarrow \text{NO}(g) + 2 \text{H}_2\text{O}(l)$
+0.80	$\text{Ag}^+(aq) + e^- \longrightarrow \text{Ag}(s)$
+0.77	$\text{Fe}^{3+}(aq) + e^- \longrightarrow \text{Fe}^{2+}(aq)$
+0.68	$\text{O}_2(g) + 2 \text{H}^+(aq) + 2 e^- \longrightarrow \text{H}_2\text{O}_2(aq)$
+0.59	$\text{MnO}_4^-(aq) + 2 \text{H}_2\text{O}(l) + 3 e^- \longrightarrow \text{MnO}_2(s) + 4 \text{OH}^-(aq)$
+0.54	$\text{I}_2(s) + 2 e^- \longrightarrow 2 \text{I}^-(aq)$
+0.40	$\text{O}_2(g) + 2 \text{H}_2\text{O}(l) + 4 e^- \longrightarrow 4 \text{OH}^-(aq)$
+0.34	$\text{Cu}^{2+}(aq) + 2 e^- \longrightarrow \text{Cu}(s)$
0 [defined]	$2 \text{H}^+(aq) + 2 e^- \longrightarrow \text{H}_2(g)$
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-0.83	$2 \text{H}_2\text{O}(l) + 2 e^- \longrightarrow \text{H}_2(g) + 2 \text{OH}^-(aq)$
-1.66	$\text{Al}^{3+}(aq) + 3 e^- \longrightarrow \text{Al}(s)$
-2.71	$\text{Na}^+(aq) + e^- \longrightarrow \text{Na}(s)$
-3.05	$\text{Li}^+(aq) + e^- \longrightarrow \text{Li}(s)$

Relationship between Gibbs Free Energy and Emf

The change in the Gibbs free energy is a measure of the spontaneity of a process at Constant T and P. So does Emf, E.

ΔG for a redox reaction can be found by using the equation

$$\Delta G = -nFE$$

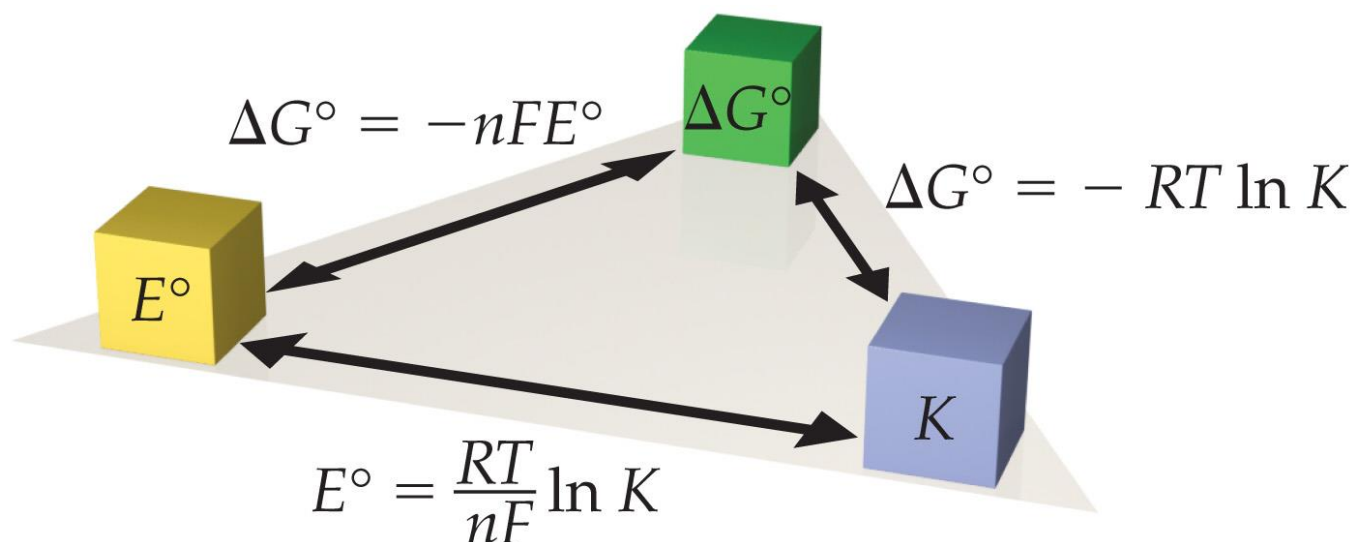
where n is the number of moles of electrons transferred, and F is **Faraday's constant**

$$1 F = 96,485 \text{ C/mol} = 96,485 \text{ J/V-mol}$$



Free Energy, Emf, and K

- How is everything related?
- **$\Delta G^\circ = -nFE^\circ = -RT \ln K$**



Sample exercise 20.10 intensive/extensive



Nernst Equation 能斯脱方程

- Remember, $\Delta G = \Delta G^\circ + RT \ln Q$
- So, $-nFE = -nFE^\circ + RT \ln Q$
- Dividing both sides by $-nF$, we get the **Nernst equation**: $E = E^\circ - \frac{RT}{nF} \ln Q$
- OR $E = E^\circ - (2.303 \frac{RT}{nF}) \log Q$
- Using **standard thermodynamic temperature (298K)** and the constants R and F ,

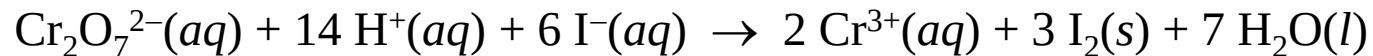
$$E = E^\circ - (0.0592\text{V}/n) \log Q$$

Equation 20.19 and Sample Exercise 20.11



Sample Exercise 20.11 Cell Potential under Nonstandard Conditions

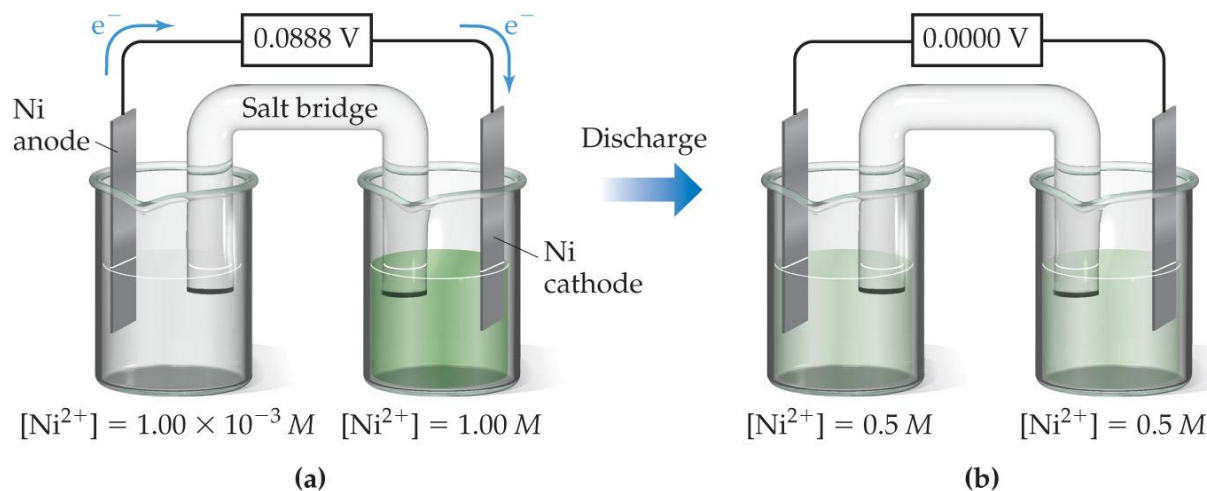
Calculate the emf at 298 K generated by a voltaic cell in which the reaction is



when $[\text{Cr}_2\text{O}_7^{2-}] = 2.0 \text{ M}$, $[\text{H}^+] = 1.0 \text{ M}$, $[\text{I}^-] = 1.0 \text{ M}$, and $[\text{Cr}^{3+}] = 1.0 \times 10^{-5} \text{ M}$.



Concentration Cells 浓差电池

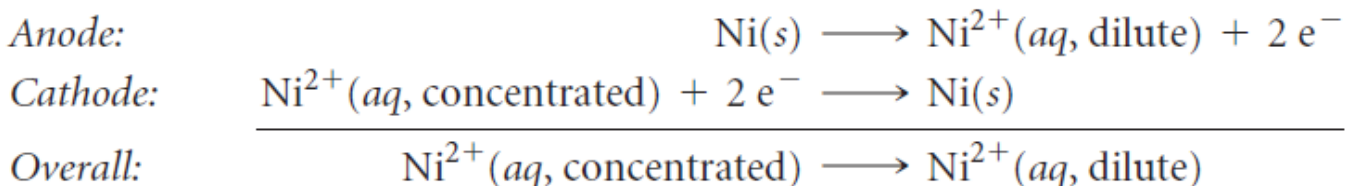
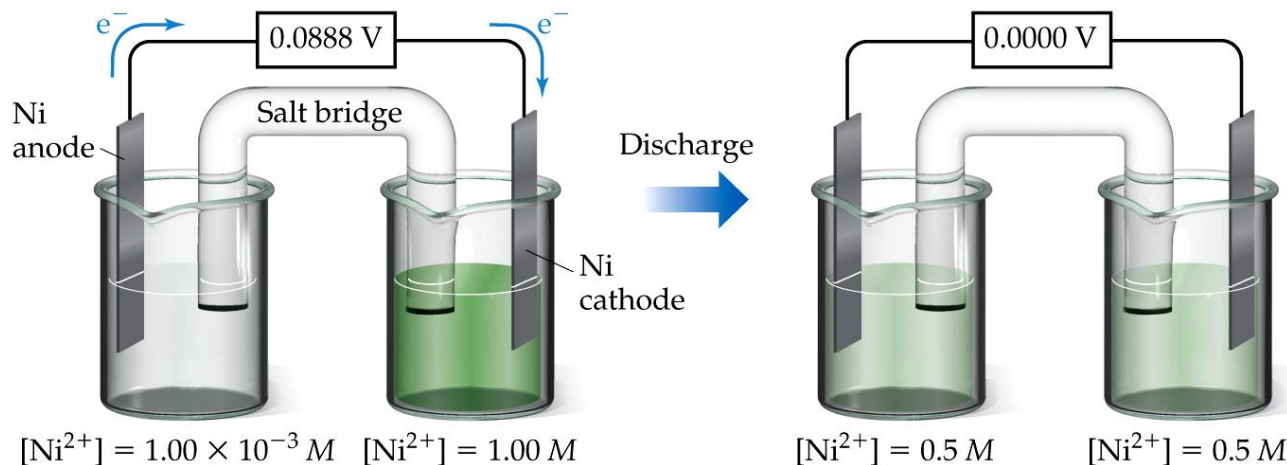


- Notice that the Nernst equation implies that a cell could be created that has the **same substance** at both electrodes, called a **concentration cell**.
- For such a cell, E_{cell}° would be 0, but Q would not.
- Therefore, as long as the concentrations are different, **E will not be 0.**

Discharge Process?

Sample Exercise 20.13

Electrochemistry



For this particular cell, we see that $n = 2$. the emf at 298 K is

$$\begin{aligned}
 E &= E^{\circ} - \frac{0.0592 \text{ V}}{n} \log Q \\
 &= 0 - \frac{0.0592 \text{ V}}{2} \log \frac{[\text{Ni}^{2+}]_{\text{dilute}}}{[\text{Ni}^{2+}]_{\text{concentrated}}} = -\frac{0.0592 \text{ V}}{2} \log \frac{1.00 \times 10^{-3} \text{ M}}{1.00 \text{ M}} \\
 &= +0.0888 \text{ V}
 \end{aligned}$$

The idea of generating a potential by **a concentration difference** is the basis for the operation of **pH meters**.

Sample Exercise 20.13 Determining pH Using a Concentration Cell

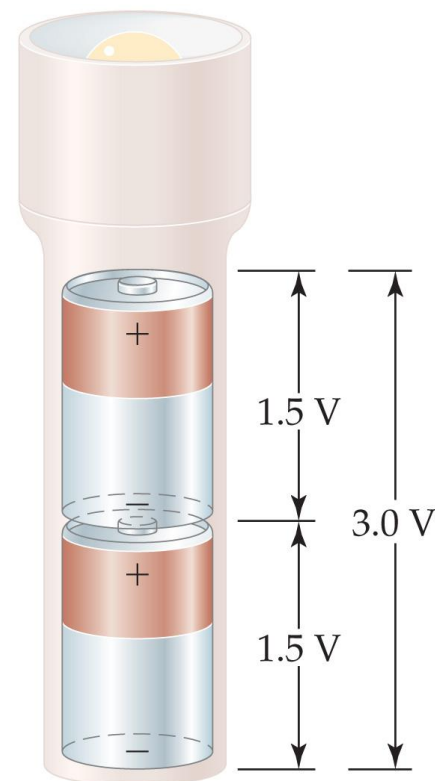
A voltaic cell is constructed with two hydrogen electrodes. Electrode 1 has $P_{\text{H}_2} = 1.00$ atm and an unknown concentration of $\text{H}^+(\text{aq})$. Electrode 2 is a standard hydrogen electrode ($P_{\text{H}_2} = 1.00$ atm, $[\text{H}^+] = 1.00 \text{ M}$). At 298 K the measured cell potential is 0.211 V, and **the electron is observed to flow from electrode 1 through the external circuit to electrode 2**. Calculate for the solution at electrode 1. What is the pH of the solution?



Some Applications of Cells

- Electrochemistry can be applied as follows:
 - ❖ **Batteries**: a portable, self-contained electrochemical power source that consists of one or more voltaic cells.
 - Batteries can be **primary cells** 原电池 (cannot be recharged when “dead”—the reaction is complete) or **secondary cells** 蓄电池 (can be recharged).
 - ❖ **Prevention of corrosion** (“rust-proofing”)
 - ❖ **Electrolysis**

Read over 20.7-20.9 if you have interest.



Electrochemistry

Some Examples of Batteries

- **Lead–acid battery 铅酸电池**: reactants and products are solids, so Q is 1 and the potential is independent of concentrations; however, made with **lead and sulfuric acid** (hazards).

PbO_2 (Cathode)/ Pb (Anode)/ H_2SO_4 (electrode)

- **Alkaline battery 碱性电池**: most common primary battery.

MnO_2 (Cathode)/ Zn (Anode)/ KOH (electrode)

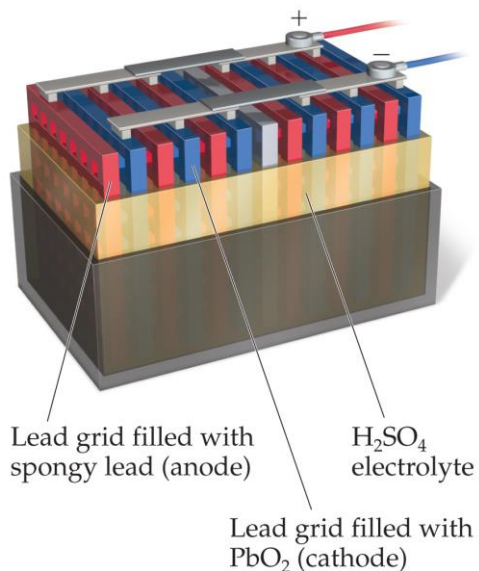
- **Ni–Cd and Ni–metal hydride batteries 镍碱性电池/ 镍氢电池** : lightweight, rechargeable; Cd is toxic and heavy, so hydrides are replacing it.

$\text{NiO}(\text{OH})_2$ (Cathode)/ Cd (Anode)/ KOH (electrode)

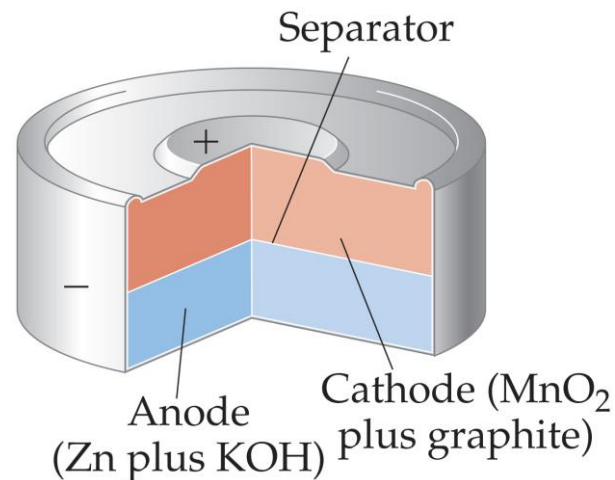
- **Lithium-ion batteries 锂电池**: rechargeable, light; produce more voltage than Ni-based batteries.



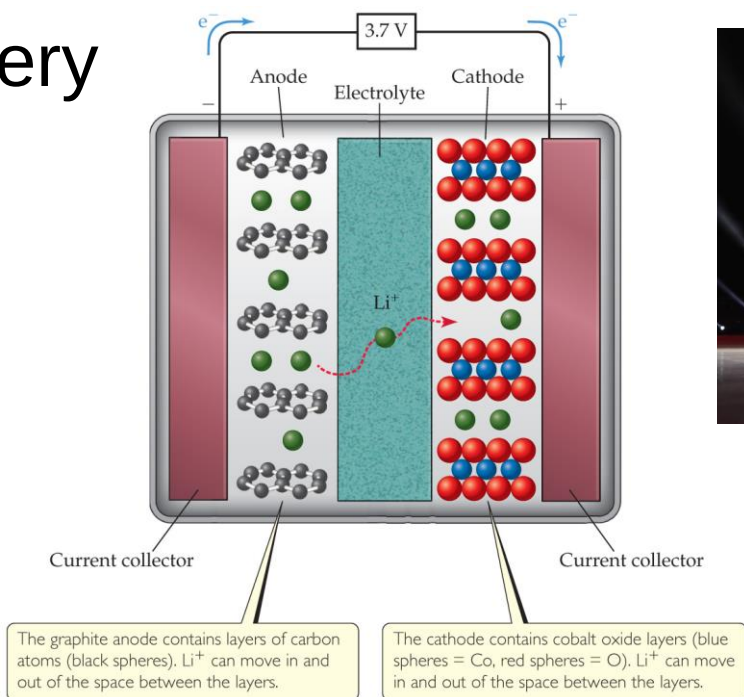
Lead–Acid Battery



Alkaline Battery



Lithium-Ion Battery

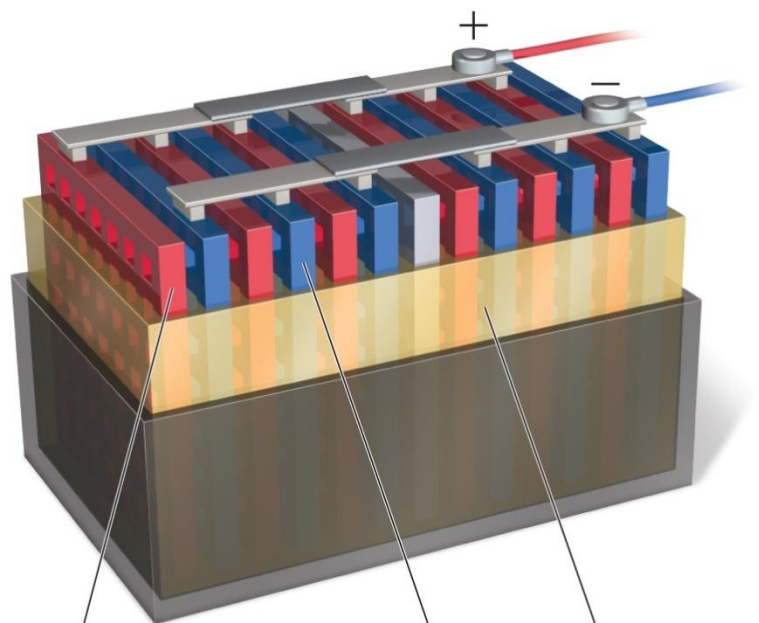


Tesla Roadster

Electrochemistry

Batteries

Lead-Acid Battery

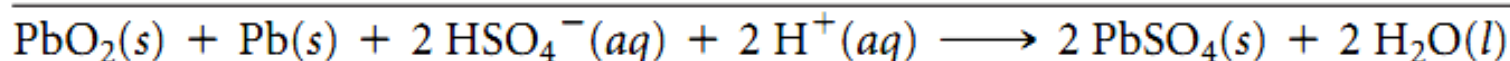
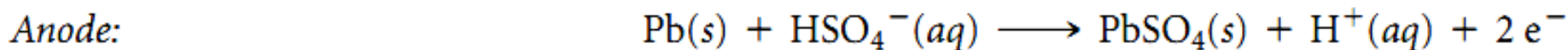
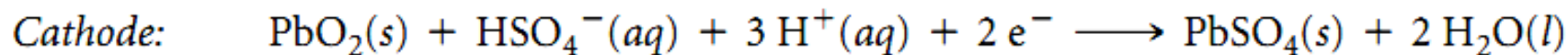


Lead grid filled with
spongy lead (anode)

H_2SO_4
electrolyte

Lead grid filled with
 PbO_2 (cathode)

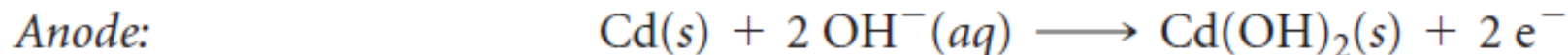
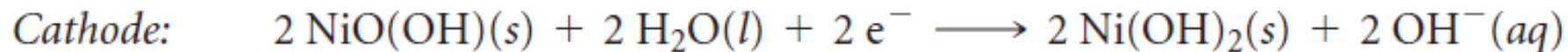
12-V automotive battery



$$E_{\text{cell}}^{\circ} = E_{\text{red}}^{\circ}(\text{cathode}) - E_{\text{red}}^{\circ}(\text{anode}) = (+1.685 \text{ V}) - (-0.356 \text{ V}) = +2.041 \text{ V}$$

Nickel-Cadmium batteries

Lightweight, readily rechargeable



Limitations:

Because cadmium is toxic, these batteries must be recycled;
Cadmium has a relatively high density, increasing weight.



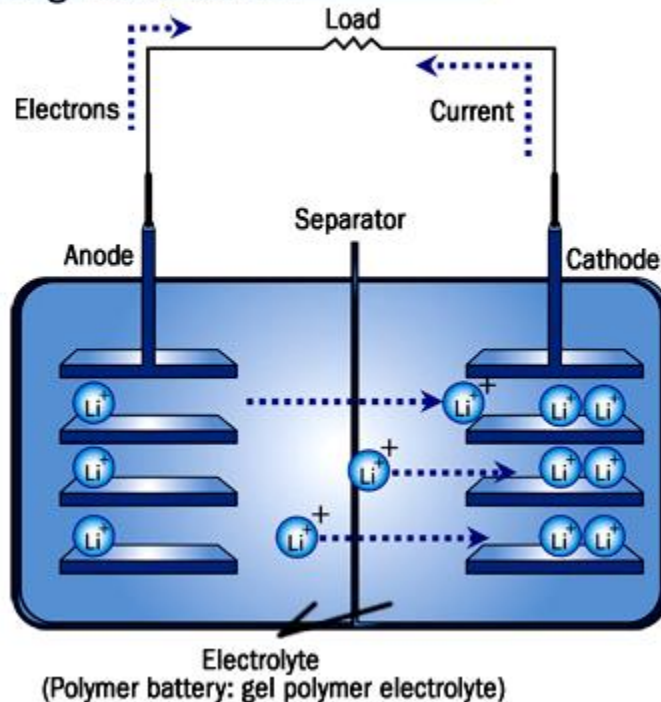
Nickel-Metal-Hydride



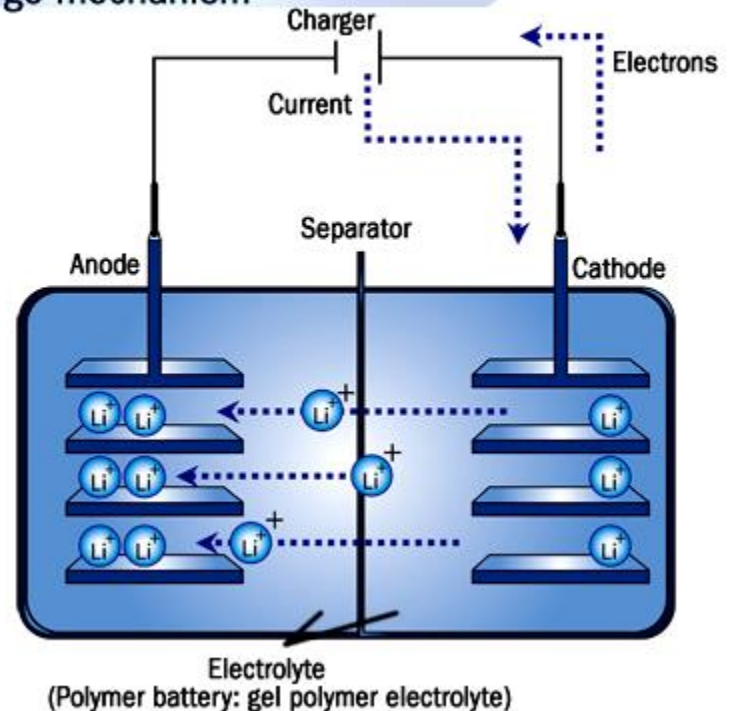
Lithium-Ion Battery

A lithium-ion battery is a family of rechargeable battery types in which **lithium ions** move from the anode to the cathode during discharge, and back when charging.

Lithium-ion rechargeable battery
Discharge mechanism



Lithium-ion rechargeable battery
Charge mechanism



Lithium-Ion Battery

In 1979, [John Goodenough](#) demonstrated a rechargeable cell with high cell voltage in the 4V range using lithium cobalt oxide (LiCoO_2) as the cathode and lithium metal as the anode.

In 1981, [Bell Labs](#) developed a workable graphite anode to provide an alternative to the lithium metal battery.



Electrochemistry



Fuel Cells

- When a fuel is burned, the energy created can be converted to electrical energy.
- Usually, this conversion is only 40% efficient, with the remainder lost as heat.
- The direct conversion of chemical to electrical energy is expected to be more efficient and is the basis for **fuel cells**.
- Fuel cells are NOT batteries; the source of energy must be continuously provided.



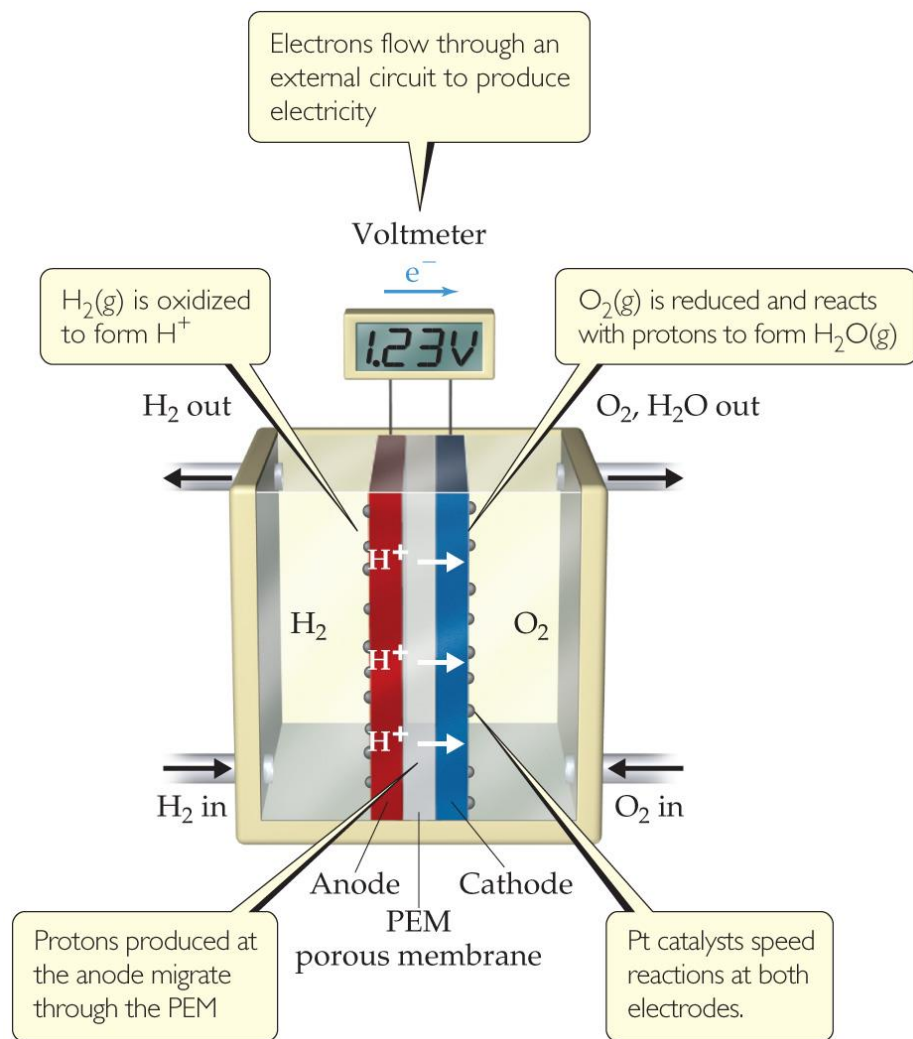
Hydrogen Fuel Cells

- In this cell, hydrogen and oxygen form water.
- The cells are twice as efficient as combustion.
- The cells use hydrogen gas as the fuel and oxygen from the air.

Toyota Mirai 500 km
Hyundai Nexo 600 km

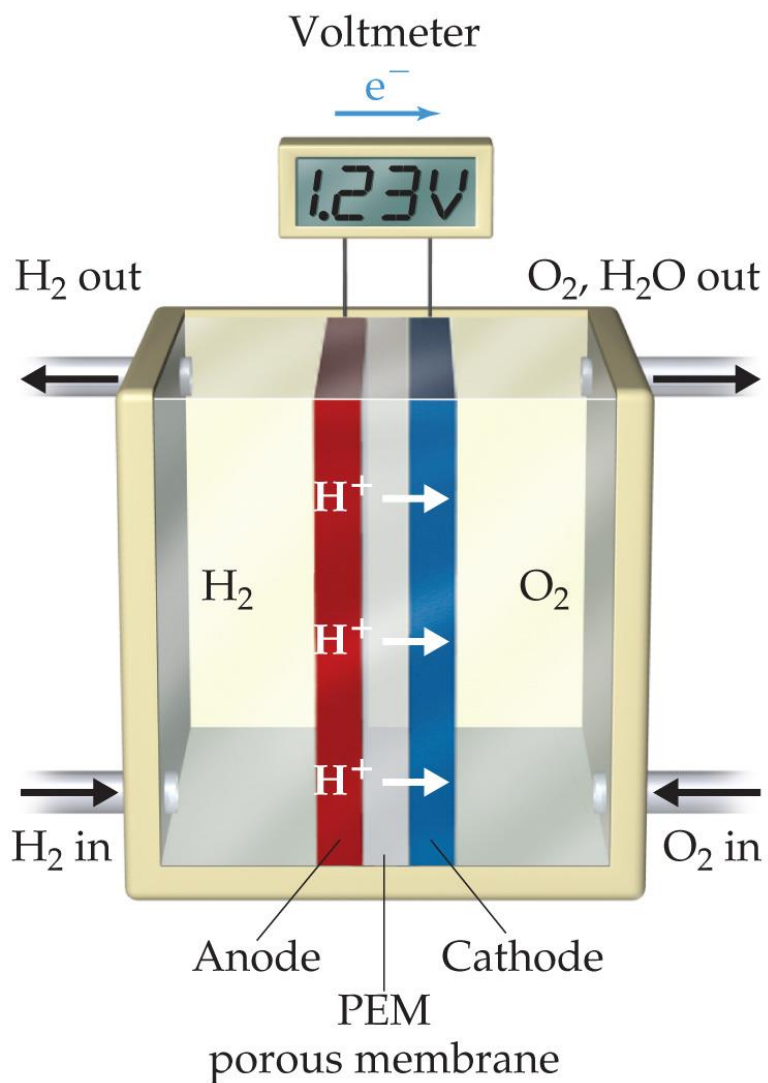


BMW Hydrogen Fuel Cell Car



Electrochemistry

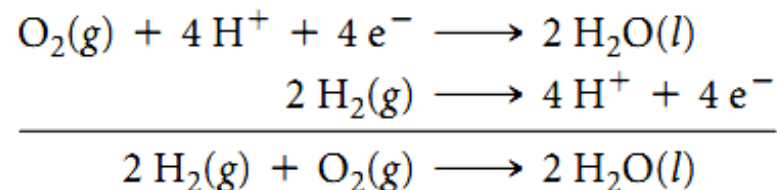
Hydrogen Fuel Cells



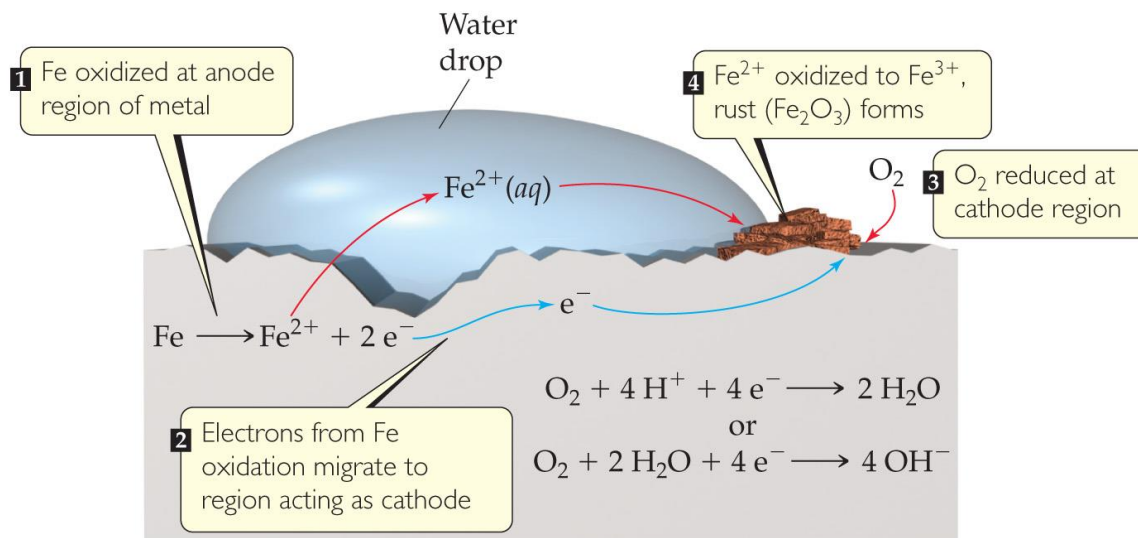
Cathode:

Anode:

Overall:

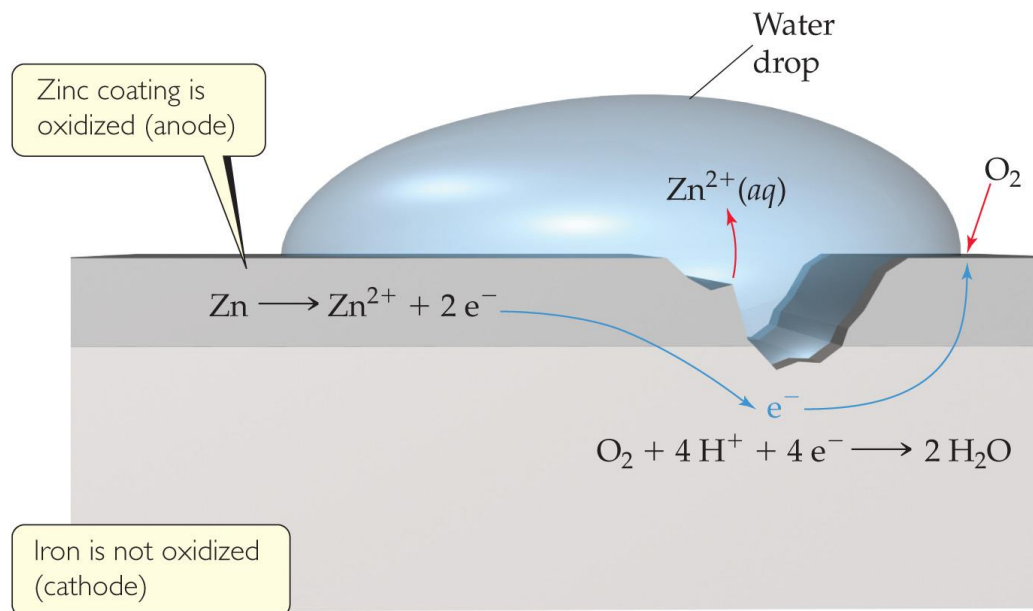


Corrosion



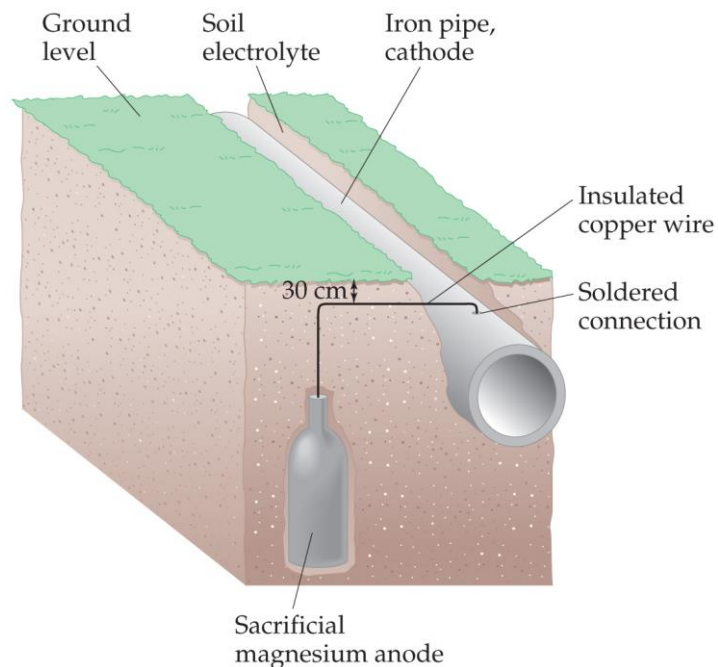
- Corrosion is oxidation.
- Its common name is *rusting*.

Preventing Corrosion



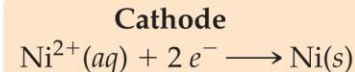
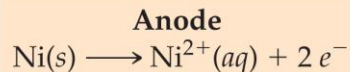
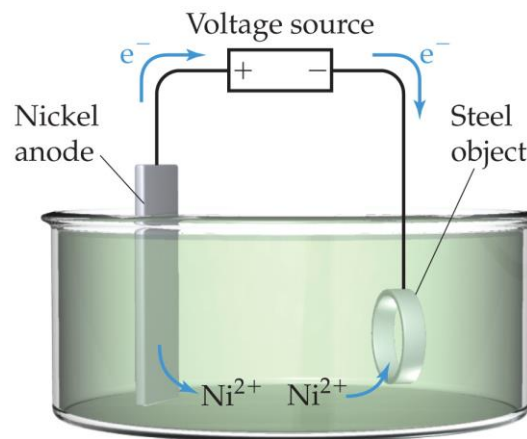
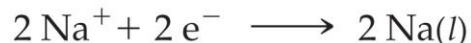
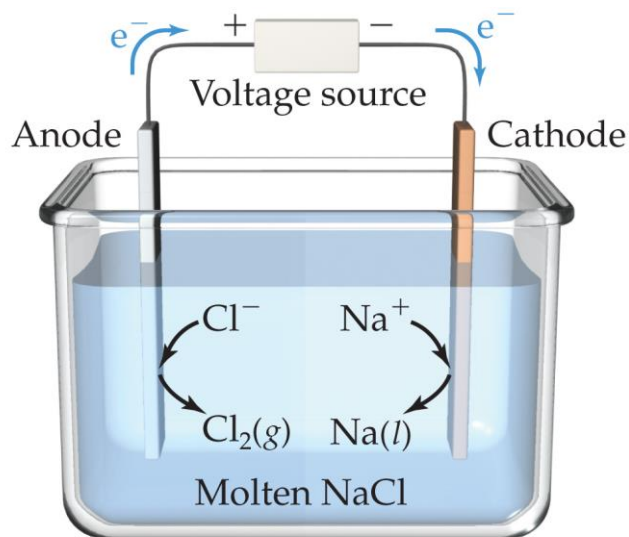
- Corrosion is prevented by coating iron with a metal that is more readily oxidized.
- **Cathodic protection** occurs when zinc is more easily oxidized, so that metal is sacrificed to keep the iron from rusting.

Preventing Corrosion



- Another method to prevent corrosion is used for underground pipes.
- A **sacrificial anode** is attached to the pipe. The anode is oxidized before the pipe.

Electrolysis



- Nonspontaneous reactions can occur in electrochemistry IF outside electricity is used to drive the reaction.
- Use of electrical energy to create chemical reactions is called **electrolysis**.

The oxidation state of nitrogen in the ammonium ion (NH_4^{1+}) is _____.

- a. +1
- b. 0
- c. -1
- d. -3

The oxidation state of nitrogen in the ammonium ion (NH_4^{1+}) is _____.

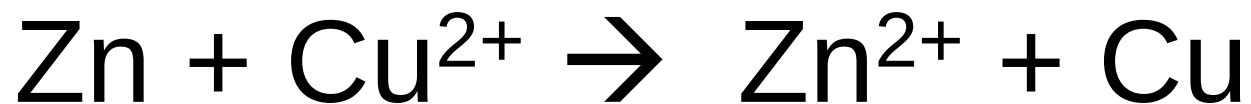
- a. +1
- b. 0
- c. -1
- d. -3**

The oxidation state of manganese in the permanganate ion (MnO_4^{1-}) is _____.

- a. -1
- b. +2
- c. +4
- d. +7

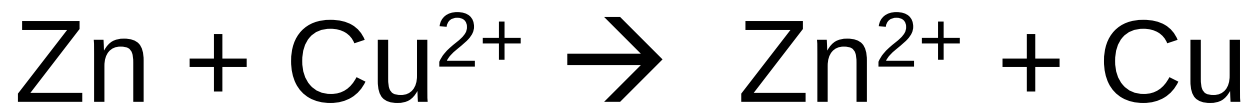
The oxidation state of manganese in the permanganate ion (MnO_4^{1-}) is _____.

- a. -1
- b. +2
- c. +4
- d. +7**



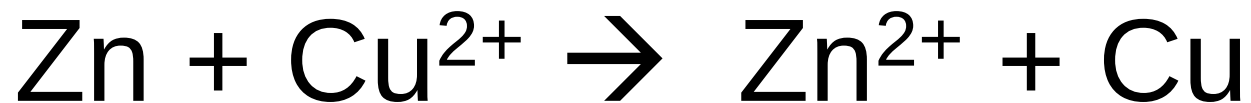
The reducing agent in the reaction above is _____.

- a. Zn
- b. Cu^{2+}
- c. Zn^{2+}
- d. Cu



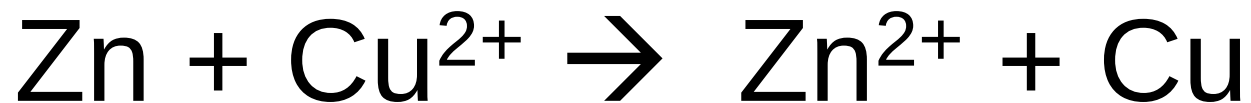
The reducing agent in the reaction above is _____.

- a. Zn
- b. Cu^{2+}
- c. Zn^{2+}
- d. Cu



The oxidizing agent in the reaction above is _____.

- a. Zn
- b. Cu^{2+}
- c. Zn^{2+}
- d. Cu



The oxidizing agent in the reaction above is _____.

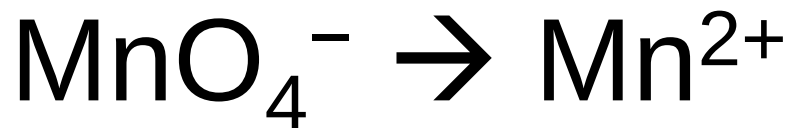
- a. Zn
- b. Cu^{2+}**
- c. Zn^{2+}
- d. Cu

To balance the half-reaction $\text{MnO}_4^- \rightarrow \text{Mn}^{2+}$ in acidic solution, _____ electrons must be added on the _____ side.

- a. 5; product
- b. 2; product
- c. 7; reactant
- d. 5; reactant

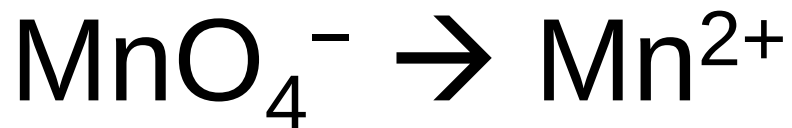
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- a. 5; product
- b. 2; product
- c. 7; reactant
- d. 5; reactant**



To balance this reaction in acidic solution, _____ H^+ must be added on the _____ side.

- a. 8; product
- b. 4; product
- c. 8; reactant
- d. 4; reactant



To balance this reaction in acidic solution, _____ H^+ must be added on the _____ side.

- a. 8; product
- b. 4; product
- c. 8; reactant**
- d. 4; reactant

If the value of the standard cell potential for a reaction is large and positive, then the reaction is

- a. at equilibrium.
- b. spontaneous.
- c. nonspontaneous.
- d. very fast.



If the value of the standard cell potential for a reaction is large and positive, then the reaction is

- a. at equilibrium.
- b. spontaneous.**
- c. nonspontaneous.
- d. very fast.



The purpose of the salt bridge in a voltaic cell is to

- a. provide H^+ ions needed to balance charges.
- b. maintain neutrality by allowing the flow of ions.
- c. serve as the site for oxidation to occur.
- d. serve as the site for reduction to occur.

The purpose of the salt bridge in a voltaic cell is to

- a. provide H^+ ions needed to balance charges.
- b. maintain neutrality by allowing the flow of ions.**
- c. serve as the site for oxidation to occur.
- d. serve as the site for reduction to occur.

Which of the following is the strongest oxidizing agent?
(If necessary, consult a table of standard reduction potentials.)

- a. F_2
- b. Cl_2
- c. Br_2
- d. I_2

Which of the following is the strongest oxidizing agent?
(If necessary, consult a table of standard reduction potentials.)

- a. F_2
- b. Cl_2
- c. Br_2
- d. I_2

Which of the following is the strongest reducing agent?
(If necessary, consult a table of standard reduction potentials.)

- a. Zn
- b. Al
- c. Na
- d. Li



Which of the following is the strongest reducing agent?
(If necessary, consult a table of standard reduction potentials.)

- a. Zn
- b. Al
- c. Na
- d. Li**





Select the true statement.

- a. Cl_2 will reduce I^- to I_2 .
- b. Cl_2 will oxidize I^- to I_2 .
- c. I_2 will reduce Cl^- to Cl_2 .
- d. I_2 will oxidize Cl_2 to Cl^- .





Select the true statement.

a. Cl_2 will reduce I^- to I_2 .

b. Cl_2 will oxidize I^- to I_2 .

c. I_2 will reduce Cl^- to Cl_2 .

d. I_2 will oxidize Cl_2 to Cl^- .



G = Gibbs free energy.

n = moles of electrons.

F = Faraday constant. E = cell potential. Change in G = ?

a. $F + nE$

b. $F - nE$

c. $-nFE$

d. $nF - E$



G = Gibbs free energy.

n = moles of electrons.

F = Faraday constant. E = cell potential. Change in G = ?

a. $F + nE$

b. $F - nE$

c. $-nFE$

d. $nF - E$



The Nernst Equation is most useful for determining cell potentials when _____ are nonstandard.

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Reduction occurs at the

- a. anode, in both galvanic and electrolytic cells.
- b. cathode, in both galvanic and electrolytic cells.
- c. anode in galvanic cells and cathode in electrolytic cells.
- d. cathode in galvanic cells and anode in electrolytic cells.



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I = current in amperes

t = time in seconds

Which is true?

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Corrosion of metals can be prevented by all of the following methods **except**

- a. a sacrificial anode.
- b. a salt bridge.
- c. formation of an oxide coating.
- d. paint.

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